

Construction of a Manifold HROM with Adaptive ECM Weights Applied to a Metamaterial Unit Cell

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Abstract

This document describes the construction of a manifold-based hyperreduced-order model with adaptive integration weights, based on the Empirical Cubature Method, and applied to a nonlinear metamaterial unit cell. The objective is to present, in a precise and reproducible manner, the complete offline procedure used to identify the reduced manifold, construct the adaptive cubature rule, regress the state-dependent weights, and assess the resulting hyperreduced approximation.

The example considered here corresponds to a bivariate homogenization problem in which the microscopic response of a metamaterial cell is parameterized by two macroscopic deformation variables. In contrast with a standard ECM rule, where a fixed set of weights is used over the whole reduced manifold, the present formulation allows the cubature weights to depend explicitly on the reduced master coordinates. The resulting approach will be referred to as the manifold adaptive-weight Empirical Cubature Method, or MAW-ECM.

1 Introduction and motivation

2 Introduction (to the metarial cell example)

The purpose of this document is to describe the procedure used to construct a manifold-based hyperreduced-order model equipped with adaptive cubature weights. The method is based on the Empirical Cubature Method (ECM), but departs from the classical setting by allowing the reduced integration weights to vary over the reduced manifold.

The problem considered is a nonlinear metamaterial unit cell subjected to a family of macroscopic deformation states. The corresponding full-order simulations provide the training data from which a reduced nonlinear manifold is constructed. This manifold is parameterized by a small number of master variables, which in the present example are associated with two independent macroscopic deformation components.

Once the reduced manifold has been identified, the objective is not merely to reduce the dimension of the displacement field, but also to reduce the cost of evaluating the homogenized response. In a standard HROM, this is achieved by replacing the full Gauss-point integration rule with a sparse ECM rule. However, a fixed cubature rule must remain accurate over the entire sampled manifold, and therefore may require more integration points than necessary for individual deformation states.

The central idea explored here is to construct a state-dependent cubature rule. More precisely, a reduced set of integration points is retained, but their associated weights are allowed to depend on the master coordinates of the manifold. In this way, the hyperreduction is adapted to the local region of the nonlinear response while preserving the structure of an ECM-type approximation.

The goal of the present document is therefore to give a detailed account of the complete construction:

- the generation and organization of the training data for the metamaterial cell;
- the construction of the reduced nonlinear manifold;
- the baseline fixed-weight ECM approximation;
- the adaptive-weight MAW-ECM construction;
- the regression of the cubature weights as functions of the master variables;
- and the final numerical assessment of the resulting HROM.

3 Problem setup

3.1 Homogenization of a nonlinear metamaterial cell

We consider a nonlinear computational homogenization problem for a metamaterial unit cell undergoing large strains and large rotations. The objective is to determine the homogenized (macroscopic) stress associated with a prescribed macroscopic deformation. This class of problems is representative of multi-query settings, in which the same microscale model must be solved for a large number of input configurations, thereby motivating the use of reduced-order models.

In two spatial dimensions, the macroscopic deformation gradient contains four independent components. In the present study, rigid-body rotations are removed, and the deformation tensor is taken to be symmetric. Moreover, in order to facilitate visualization and interpretation, only two of the three independent components are retained as input parameters. This allows us to represent the latent space explicitly and to analyze, in a graphical manner, the dependence of the reduced quantities—most notably, the adaptive integration weights—on the macroscopic state.

The material behavior is assumed to be hyperelastic, following a Neo-Hookean constitutive law. As a consequence, the problem is not history-dependent, and there exists a unique mapping between the macroscopic deformation and the homogenized stress. This fact might suggest the possibility of constructing a direct regression between input strains and output (average) stresses. However, such an approach is deliberately not pursued here.

The objective of the present work is fundamentally different. Rather than approximating input-output relations at the macroscopic level, we seek a compressed representation of the full mechanical state of the system. In particular, the methodology is designed to reconstruct not only homogenized quantities, but also the underlying microscopic displacement fields. This capability is essential in more advanced multiscale frameworks, especially those based on domain decomposition, where the consistent transfer of kinematic fields across scales plays a central role.

Accordingly, the emphasis is placed on the construction of a physically consistent reduced-order model, in which the only approximations stem from the restriction of the solution to a low-dimensional manifold and from the hyper-reduction of the associated integrals. Within this setting, the primary goal of this work is to investigate in depth the construction of adaptive cubature rules.

To this end, the methodology adopted preserves the structure of the underlying boundary value problem. The only approximations introduced are: (i) the restriction of the displacement field to a low-dimensional nonlinear manifold, and (ii) the approximation of spatial integrals through a reduced cubature rule. These are the same types of approximations that arise in standard Galerkin procedures, and therefore do not compromise the physical consistency of the model.

The ultimate objective of this example is to analyze the behavior of the adaptive-weight ECM algorithm in a controlled setting. For this reason, the regression component is intentionally

kept simple, so as to isolate and better understand the influence of the adaptive weights. The emphasis is thus placed on the structure and properties of the resulting hyperreduction, rather than on the use of sophisticated data-driven fitting techniques.

3.1.1 On the use of first-order homogenization

A word of clarification is in order. The use of first-order computational homogenization for the present metamaterial cell should not be interpreted as an endorsement of its validity. For microstructures exhibiting strong internal mechanisms (e.g., buckling), the assumptions of scale separation may be questionable, and higher-order or micromorphic frameworks are often more appropriate (see, e.g., [5]).

Its use here is purely illustrative. The objective is to provide a controlled setting in which the adaptive-weight ECM can be analyzed in detail. The proposed methodology is not restricted to first-order schemes and can be extended to more general multiscale formulations —of course, at the expense of increasing the number of input parameters.

3.2 Metamaterial cell geometry

The unit cell considered in this work is inspired by curved-beam negative-stiffness metamaterials such as those proposed in [8]. These configurations originate from the classical curved-beam bistable mechanisms introduced by Qiu et al. [17], and were later extended to cellular architectures exhibiting negative stiffness and enhanced energy absorption capabilities [9]. The resulting macroscopic response is governed primarily by geometric nonlinearities associated with snap-through instabilities of slender curved members.

Following [8], the curved portions of the cell are described by a cosine-type profile. In particular, the beam centerline is parameterized as

$$w(x) = \frac{h}{2} \left[1 - \cos \left(\frac{\pi x}{L} \right) \right],$$

where L denotes the span of the beam and h its amplitude. This representation provides a smooth geometry with controlled curvature, known to facilitate snap-through behavior under compressive loading.

The resulting unit cell is depicted in Fig. 1. It consists of a curved ligament confined between two lateral walls and connected through a transverse member. The curved beam acts as the primary nonlinear mechanism, while the surrounding structure enforces kinematic constraints that shape the global response. This simplified configuration retains the essential features of mechanism-based metamaterials while remaining suitable for a controlled study of the adaptive-weight ECM framework.

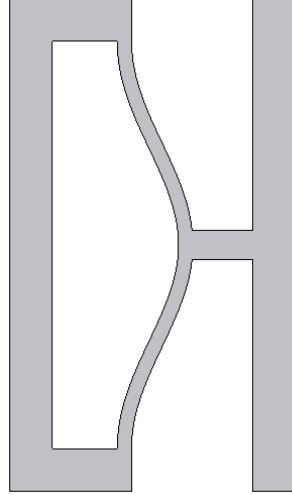


Figure 1: Curved-beam metamaterial unit cell considered in this work. The geometry is inspired by negative-stiffness designs based on cosine-shaped curved beams undergoing snap-through instabilities.

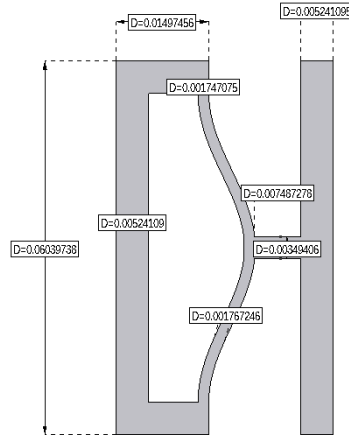


Figure 2: Geometrical definition of the curved-beam metamaterial unit cell. The characteristic dimensions of the structure are indicated in the figure. These include the overall height and width of the cell, the thickness of the lateral walls, the width of the transverse connector, and the geometric parameters defining the curved ligament. The values reported correspond to the reference configuration used in the numerical simulations (units in meters).

3.2.1 Material properties

3.2.2 Constitutive model

The base material of the metamaterial cell is modeled as an isotropic compressible Neo–Hookean solid. The elastic parameters are taken as $E = 1628$ MPa and $\nu = 0.4$, where E is Young’s modulus and ν is Poisson’s ratio (these values correspond to a high-impact ABS material).

The material parameters are defined in `ElasticMaterialDATA.NH.m`. The constitutive model is selected through `DATA.TYPE.CONSTITUTIVE.MODEL.ALL = 'NeoHookean'`. The parameters used are `E = 1628` MPa and `nu = 0.4`.

3.3 Finite element discretization

The unit cell is discretized using standard two-dimensional quadrilateral finite elements with linear interpolation (four-node elements). A characteristic element size of the order of 10^{-4} m is employed, resulting in a mesh comprising 1315 elements and 1602 nodes.

The mesh is constructed to ensure compatibility along opposite boundaries of the unit cell. In particular, a structured nodal distribution is imposed along the boundary lines so that nodes on opposing sides are aligned and can be directly paired. This requirement stems from the imposition of periodic boundary conditions in the homogenization framework.

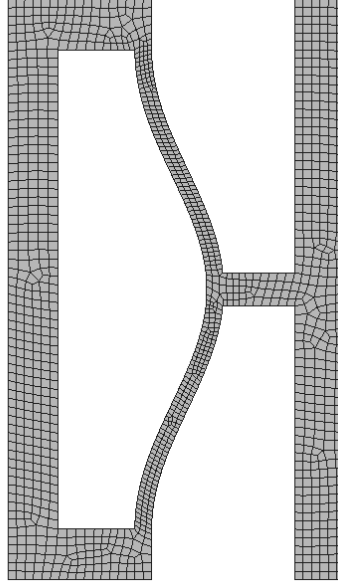


Figure 3: Finite element discretization of the metamaterial unit cell.

3.4 Boundary conditions

Since a first-order computational homogenization setting is adopted, the macroscopic input is the deformation gradient, denoted by F_{macro} . In plane strain, this tensor can be expressed as

$$F_{\text{macro}} = \begin{bmatrix} F_x & F_{xy} \\ F_{yx} & F_y \end{bmatrix}.$$

In general, this representation involves four independent components. However, part of this information corresponds to rigid-body rotations, which are not relevant for the present objective. In order to exclude rotational effects, we restrict attention to symmetric deformations by enforcing $F_{xy} = F_{yx}$. As a result, only the deformational part of F_{macro} is retained, reducing the parametric space to three independent variables.

In this example, we do not explore the complete three-dimensional deformation space. This choice is twofold. First, the purpose of the example is to analyze the adaptive-weight ECM construction, not to address the general problem of high-dimensional regression. Second, for the present cell, the dominant mechanism is activated by deformation along the weak direction of the microstructure, namely the horizontal x -direction, along which the curved beams undergo snap-through-like deformation.

Accordingly, the prescribed macroscopic deformation is restricted to two components: stretching/compression in the x -direction and in-plane shear. The former is the principal loading mode for which the negative-stiffness mechanism is designed. The latter is included because, although these devices are primarily conceived to operate under compression or tension, shear components

may naturally arise from misalignment or non-ideal loading conditions. In particular, shear perturbations may break the symmetry of the response and induce non-synchronous buckling of the curved beams, an effect that will be illustrated later.

3.5 Sampling of the macroscopic parameter space

The training set is generated by sampling a two-dimensional subspace of the macroscopic deformation space. Rather than working directly with the deformation gradient, we prescribe the macroscopic displacement gradient

$$\mathbf{G}^{\text{macro}} = \mathbf{F}_{\text{macro}} - \mathbf{I}.$$

In the present example, only two components are varied:

$$G_x^{\text{macro}} \quad \text{and} \quad G_{xy}^{\text{macro}},$$

while the transverse component is kept fixed, $G_y^{\text{macro}} = 0$. The imposed macroscopic displacement gradient therefore has the form

$$\mathbf{G}^{\text{macro}} = \begin{bmatrix} G_x^{\text{macro}} & G_{xy}^{\text{macro}} \\ G_{xy}^{\text{macro}} & 0 \end{bmatrix}, \quad \mathbf{F}_{\text{macro}} = \mathbf{I} + \mathbf{G}^{\text{macro}}.$$

The condition $G_{xy}^{\text{macro}} = G_{yx}^{\text{macro}}$ removes the rotational component and restricts the sampling to purely deformational states.

The sampled domain is defined by prescribing the lower and upper bounds of these two parameters. In the reference case considered here, the compression/stretching component along the soft direction of the cell is varied in the interval

$$G_x^{\text{macro}} \in [-0.42, 0.42],$$

whereas the shear component is varied in

$$G_{xy}^{\text{macro}} \in [-0.05, 0.05].$$

A structured grid is then introduced over this rectangular parameter domain, with 230 sampling values in the G_x^{macro} -direction and 20 sampling values in the G_{xy}^{macro} -direction.

Although the final dataset is interpreted as a structured grid in parameter space, the finite element simulations are not generated as independent isolated points. Instead, the grid is traversed through two continuous zig-zag loading trajectories. The first trajectory starts from the origin and sweeps the region with non-negative shear; the second trajectory also starts from the origin and sweeps the region with non-positive shear. The origin is therefore shared by both paths but is retained only once in the final snapshot ordering.

This construction has two advantages. First, the finite element states are generated along continuous loading paths, which is convenient for nonlinear quasi-static simulations. Second, every snapshot is assigned a unique position in the structured parameter grid. This makes it possible to reshape reduced quantities, such as master coordinates, slave coordinates, or adaptive weights, back onto the two-dimensional parameter domain for visualization and regression.

The numerical experiment is launched from `Run_HOMOG_2p.m`. The input data are defined in `DATAINPUTS/Data_M3.m`. The boundary-condition sampling is specified in `BC2v-M.m`, where the parameter bounds are `ParameterDomain.F.11 = [-0.42, 0.42]`, `ParameterDomain.F.12 = [-0.05, 0.05]`, with `nsteps.F.11 = 230` and `nsteps.F.12 = 20`. The trajectories and the structured snapshot ordering are generated by `DetermineTrajectories_cg.m`, which calls `buildGridTrajAndLabels.m`.

The sampled macroscopic parameter domain is shown in Fig. 4. The two varying input quantities are the components of the macroscopic displacement gradient, namely G_x^{macro} and G_{xy}^{macro} . The former controls the horizontal stretching/compression of the unit cell, while the

latter introduces a symmetric shear perturbation. The domain is intentionally restricted to a two-dimensional rectangle in order to keep the regression problem simple and to allow a direct graphical inspection of the adaptive weights.

The lower bound $G_x^{\text{macro}} = -0.42$ corresponds to a highly compressed configuration in which the curved ligaments approach the opposite wall, without reaching self-contact in the reference simulations. The corner configurations shown in the figure illustrate the extreme combinations of axial stretching/compression and positive/negative shear. The two colored paths represent the loading trajectories used to traverse the grid of sampling points.

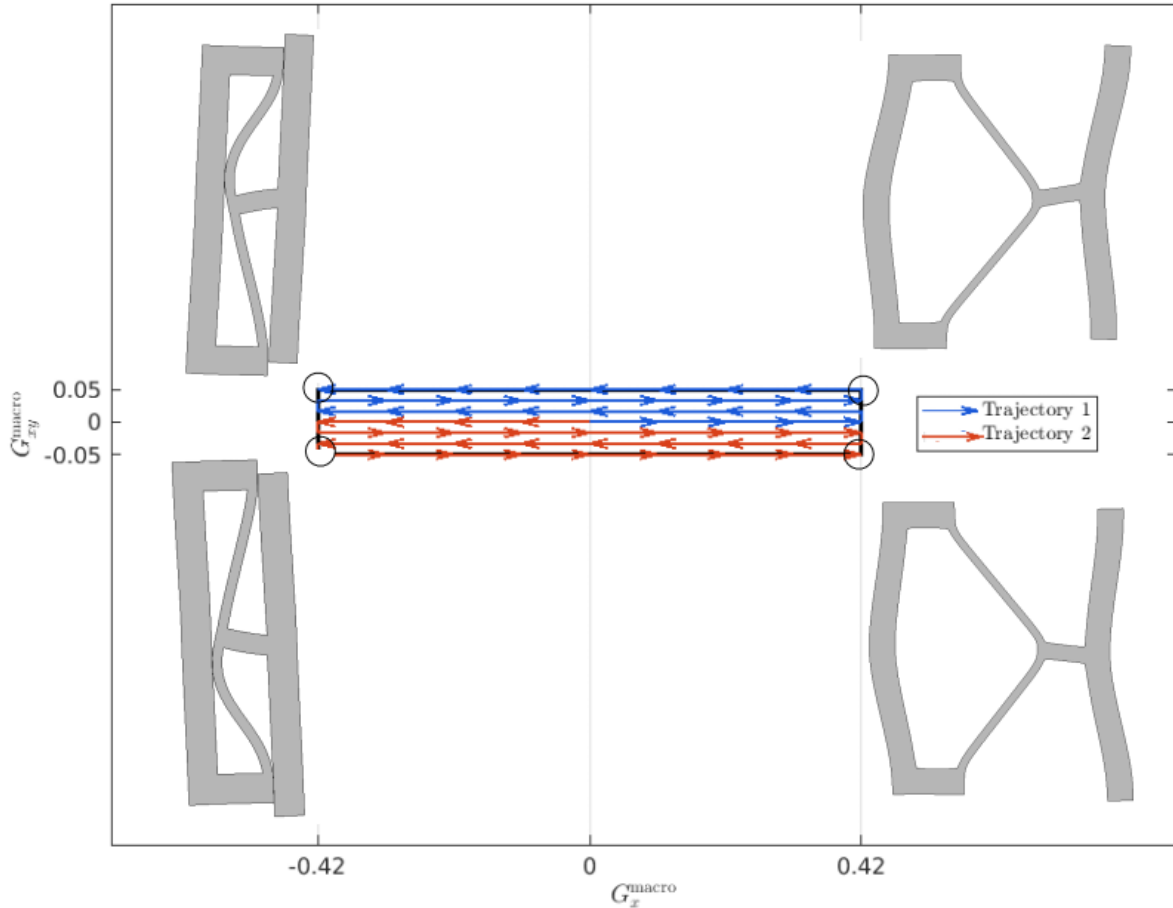


Figure 4: Sampled two-dimensional macroscopic parameter domain. The varying input quantities are the components of the macroscopic displacement gradient, G_x^{macro} and G_{xy}^{macro} , while the transverse component is kept fixed. The rectangular domain is bounded by the minimum and maximum values used in the training set. The deformed configurations at the four corners illustrate the extreme combinations of horizontal compression/stretching and positive/negative shear. The two colored curves indicate the loading trajectories used to traverse the parameter grid.

4 Finite element formulation of the unit-cell problem

4.1 Microscopic equilibrium problem

For each prescribed macroscopic displacement gradient $\mathbf{G}^{\text{macro}}$, the state of the unit cell is determined by the microscopic displacement field, represented after finite element discretization by the nodal vector $\mathbf{d}$. We work directly with the *total displacement field*, without separating fluctuations.

Under periodic boundary conditions, the displacement vector admits the decomposition

$$\mathbf{d} = \mathbf{A} \mathbf{d}_L + \bar{\mathbf{U}} \boldsymbol{\mu}. \quad (1)$$

Here, $\mathbf{d}_L$ collects the independent unknowns of the problem, namely the interior degrees of freedom together with a subset of boundary¹ degrees of freedom chosen as masters (in the present case, nodes on the left boundary and on the bottom edge, excluding corner nodes). The matrix $\mathbf{A}$ is the constraint-lifting operator associated with the periodic boundary conditions: it maps the independent unknowns $\mathbf{d}_L$ into the full nodal displacement space and contains the master–slave coupling imposed by periodicity. The remaining boundary nodes (top and right edges, as well as corners) are treated as slave degrees of freedom.

The matrix $\bar{\mathbf{U}}$ encodes the affine macroscopic deformation. Its rows are zero for all independent degrees of freedom (interior and master nodes), while nonzero entries appear only for slave boundary nodes and corner nodes, where the displacement is prescribed through the macroscopic gradient.

The vector $\boldsymbol{\mu}$ parameterizes the imposed macroscopic deformation; in the present setting, $\boldsymbol{\mu} = (G_x^{\text{macro}}, G_{xy}^{\text{macro}})^T$, with $G_y^{\text{macro}} = 0$.

For a node $\mathbf{X} = (X, Y)^T$ belonging to a slave boundary, the affine macroscopic contribution is defined in the reference configuration as $\bar{\mathbf{u}}(\mathbf{X}) = \mathbf{G}^{\text{macro}} \mathbf{X}$, which in the present case yields $\bar{\mathbf{u}}(\mathbf{X}) = (G_x^{\text{macro}} X + G_{xy}^{\text{macro}} Y, G_{xy}^{\text{macro}} X)^T$. This contribution is assembled into $\bar{\mathbf{U}} \boldsymbol{\mu}$, acting only on constrained boundary degrees of freedom.

The microscopic equilibrium is expressed through the finite element nodal residual, which in the present case (with no nodal external forces) reduces to the nodal vector of internal force vector:

$$\mathbf{f}_{\text{int}}(\mathbf{d}) = \int_{\Omega_0} \mathbf{B}^T(\mathbf{X}) \mathbf{P}(\mathbf{d}, \mathbf{X}) d\Omega_0. \quad (2)$$

After finite element discretization and Gauss quadrature, the above adopts the expression

$$\mathbf{f}_{\text{int}}(\mathbf{d}) = \sum_{e=1}^{n_{\text{el}}} \mathbf{L}_e^T \left(\sum_{g=1}^{n_{\text{gp}}^{(e)}} \mathbf{B}_{e,g}^T \mathbf{P}_{e,g} w_{e,g} \right), \quad (3)$$

where $\mathbf{L}_e$ is the standard Boolean assembly operator mapping element degrees of freedom to global ones, $\mathbf{B}_{e,g}$ is the element-level displacement-gradient operator (Voigt notation), $\mathbf{P}_{e,g}$ is the first Piola–Kirchhoff stress and $w_{e,g}$ denotes the physical quadrature weight associated with Gauss point g of element e , including both the parent-domain Gauss weight and the Jacobian determinant of the isoparametric mapping.

The above expression corresponds to the standard finite element nodal internal force vector. However, the equilibrium problem is posed in the space of independent degrees of freedom through the lifting operator $\mathbf{A}$, in accordance with the principle of virtual work. This leads to the constrained equilibrium condition $\mathbf{A}^T \mathbf{f}_{\text{int}}(\mathbf{d}) = \mathbf{0}$. Using (3), and reindexing the element/Gauss-point pairs (e, g) by a single global index, this can be written as

$$\sum_{g=1}^{n_{\text{gp}}} \mathbf{r}_g^{\text{FE}}(\mathbf{d}) w_g^{\text{FE}} = \mathbf{0}, \quad (4)$$

where

$$\mathbf{r}_g^{\text{FE}}(\mathbf{d}) := \mathbf{A}^T \mathbf{L}_e^T \mathbf{B}_{e,g}^T \mathbf{P}_{e,g}(\mathbf{d}) \in \mathbb{R}^{n_L} \quad (5)$$

denotes the internal force contribution (per unit volume) associated with the global Gauss point g , and $w_g^{\text{FE}} > 0$ is the corresponding quadrature weight (here, the superscript “FE” emphasizes that w_g^{FE} denotes the quadrature weights of the underlying finite element discretization,

¹Here and in what follows “boundary” refers to the external boundary $\partial\Omega_0$ of the reference unit cell, i.e., the interface with adjacent cells in the periodic homogenization setting.

to be distinguished later from the cubature weights used in hyperreduction.). Eq. 4 constitutes the full-order nonlinear problem used to generate the snapshot data; subsequent model reduction acts on $\mathbf{d}_L$, i.e., on the independent microscopic degrees of freedom after enforcement of periodicity and affine lifting.

Remark 1 (Gauss-point vs element-based representation). *An equivalent expression can be written at the element level by regrouping the Gauss-point contributions within each element. In that case, the summation runs over elements, the weights are replaced by the element measure, and the corresponding contribution is defined as the element-averaged internal force. Equivalently, one may incorporate the quadrature weights into the definition of $\mathbf{r}_g^{\text{FE}}$, so that the equilibrium equation is expressed as a sum of (sparse) elemental internal force contributions.*

The general infrastructure for imposing time-dependent Dirichlet and periodic boundary conditions is implemented in `DirichletCONDtime_GENERAL.m`. In the present case, periodic homogenization boundary conditions are imposed through `DirichletCONDtime.PeriodHomogQ4samALL.m`, which calls `PeriodicQ4_homogCAN.m`. For comparison, the homogeneous affine Dirichlet version is documented in `DirichletCONDtime_homogZEROall.m`.

4.2 Homogeneized stresses

Although the reduced model reconstructs the microscopic displacement field, the primary quantity of interest in first-order computational homogenization is the homogenized first Piola–Kirchhoff stress. This quantity can be obtained either as the volume average of the microscopic stress field or, equivalently, from the boundary reactions [15]. Using the volume-average representation, we write:

$$\mathbf{P}^{\text{macro}} = \frac{1}{|\Omega_0|} \int_{\Omega_0} \mathbf{P}(\mathbf{d}, \mathbf{X}) d\Omega_0 = \frac{1}{|\Omega_0|} \sum_{g=1}^{n_{\text{gp}}} \mathbf{P}_g(\mathbf{d}) w_g^{\text{FE}}. \quad (6)$$

In the finite element post-processing, the reaction-based homogenized stresses are evaluated in `CHECK_consistency_non_gen3Rg.m`. When `ComputeHomogeneizedStresses = 1`, the code collects all boundary nodes, builds the matrix `X_homog_stress` from their reference coordinates, computes `PK1stress_homg = X_homog_stress*REACTIONS_FE(DOFsBND,:)`, and normalizes by `VOL = sum(OPERFE.wSTs)`.

5 Standard hyperreduced-order model

5.1 Linear reduced subspace

As a first compressed representation, we consider a standard projection-based ROM in which the independent displacement vector $\mathbf{d}_L$ is approximated in a linear reduced subspace,

$$\mathbf{d}_L \approx \mathbf{\Phi} \mathbf{q}, \quad (7)$$

where $\mathbf{\Phi} \in \mathbb{R}^{n_L \times r}$ contains precomputed displacement modes, $\mathbf{q} \in \mathbb{R}^r$ is the vector of reduced coordinates, n_L is the number of independent degrees of freedom after imposing periodic constraints, and r is the dimension of the reduced space. In the present problem, $n_L = 3018$.

The basis $\mathbf{\Phi}$ is obtained from the snapshot matrix

$$\mathbf{D}_L = [\mathbf{d}_L(\mu_1) \quad \mathbf{d}_L(\mu_2) \quad \cdots \quad \mathbf{d}_L(\mu_{n_s})] \in \mathbb{R}^{n_L \times n_s}, \quad (8)$$

where n_s denotes the number of training snapshots. Since the parameter domain is sampled on a structured two-dimensional grid, the total number of snapshots is $n_s = 4852$, stored in two trajectory blocks of size 3018×2426 after restriction to the independent degrees of freedom.

The modal basis is computed by a truncated singular value decomposition of $\mathbf{D}_L$. In practice, this operation may be performed by any equivalent memory-aware SVD strategy, including block, partitioned, or randomized variants. In the present implementation we use the sequential

randomized SVD machinery described in [12]. The truncation criterion is based on the relative Frobenius norm of the discarded singular values.

The standard ROM basis is generated by running `Run.HOMOG_2p.m` with input file `DATAINPUTS/Data.M3std.m`, where `DATAoffline.Hyperreduction.METHOD.proj` = 'STANDARD' and `DATA.interp.METHOD.SELECT-REFERENCE.MODE` = 'STANDARD-ROM'. The offline construction is performed in `Determine.qinf.qsup.STDROM.m`. The snapshot blocks are first restricted to DOF1; in the present case `SNAPdisp` = {3018 x 2426, 3018 x 2426}. With `DATAoffline.errorDISP` = 1e-6, the call `[Phi,S,V] = SRSVD(SNAPdisp,TOL_BLOCK,DATAsvd)` returns `size(Phi,2)` = 52.

5.2 Compressibility of the displacement snapshots

Figure 5 shows the relative Frobenius truncation error obtained from the singular values of D_L . The decay is regular, but not sufficiently steep to regard the solution set as strongly compressible by a very low-dimensional linear subspace.

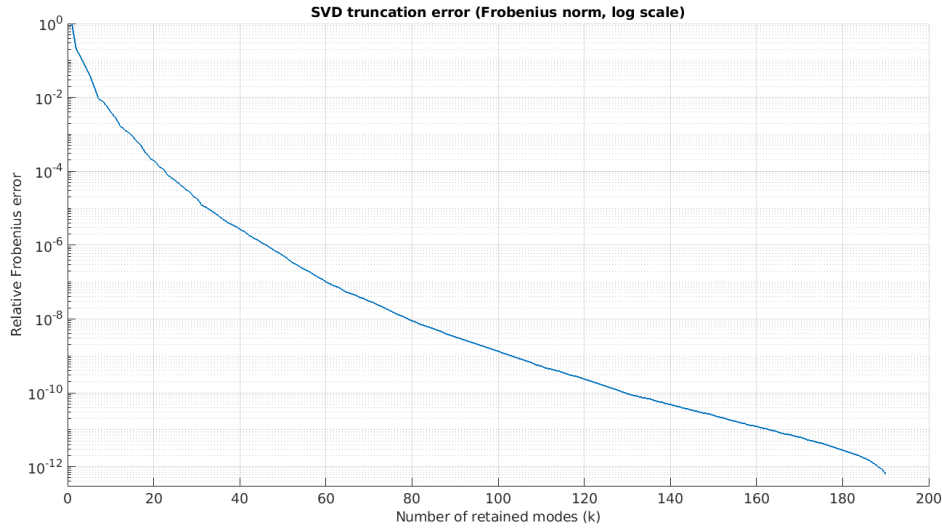


Figure 5: Relative Frobenius truncation error associated with the singular values of the displacement snapshot matrix D_L .

Indeed, even if we have only two macroscopic variables, a moderate level of accuracy already requires a relatively large number of modes. For instance, for a relative Frobenius truncation error of order 10^{-6} we need $r = 52$ for the problem under consideration (for illustration purposes, some of these modes are shown in Figure ??). Reducing the tolerance to 10^{-5} and 10^{-4} decreases the number of retained modes to $r = 37$ and $r = 25$, respectively.

The implications of these truncation levels are not immediately apparent from the a priori SVD error alone. A more meaningful assessment is obtained by examining the a posteriori errors in the reconstructed fields and derived quantities, such as the homogenized stresses. This analysis is deferred to Section 5.6, where the impact of the reduced basis dimension on both displacement and stress accuracy is quantified.

Remark 2. *This behavior is a manifestation of the classical Kolmogorov n -width limitation of linear reduced spaces: the solution manifold may be low-dimensional in terms of its parametrization but poorly aligned with any low-dimensional affine or linear subspace. The situation is expected to deteriorate further when additional macroscopic deformation components are varied. In particular, including the transverse strain component would increase the complexity of the displacement manifold, and this effect would be even more pronounced in three-dimensional settings, where up to six independent strain components may act as inputs. Consequently, the standard ROM provides a useful baseline, but it is not the most appropriate reduced representation for the present problem.*

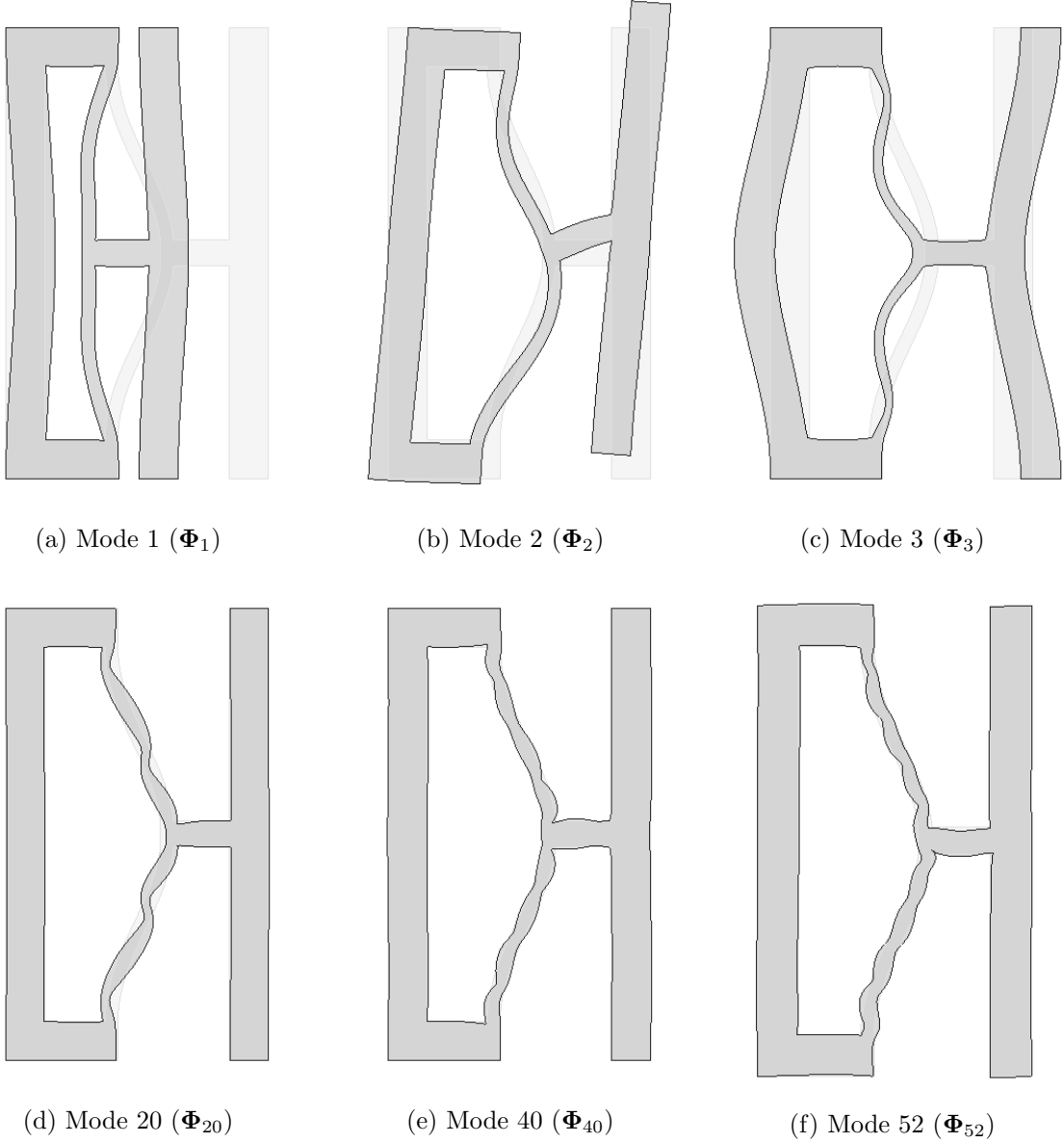


Figure 6: Deformed shapes associated with selected modes of the ROM basis. Although the modes are defined only on the independent degrees of freedom, they are extended to the full displacement field for visualization purposes through the constraint-lifting operator. The first modes (1–3) capture global deformation patterns associated with the macroscopic response, whereas higher modes (20, 40, 52) are increasingly localized and mainly describe deformation and buckling mechanisms in the curved ligaments.

5.3 Equilibrium equation in the reduced space and hyperreduction

We take as starting point the constrained equilibrium equation written in Gauss-point form, cf. (4). Introducing the linear ROM approximation and restricting the virtual displacements to the same space, the Galerkin form of the principle of virtual work yields

$$\sum_{g=1}^{n_{\text{gp}}} \mathbf{r}_g^\Phi(\mathbf{q}; \boldsymbol{\mu}) w_g^{\text{FE}} = \mathbf{0}, \quad (9)$$

where

$$\mathbf{r}_g^\Phi(\mathbf{q}; \boldsymbol{\mu}) := \boldsymbol{\Phi}^T \mathbf{r}_g^{\text{FE}} \in \mathbb{R}^r. \quad (10)$$

Equation (9) defines a system of r nonlinear equations for the reduced coordinates $\mathbf{q}$. It is important to note that, although the equilibrium equation has been modified through the projection onto the reduced space (as reflected in $\mathbf{r}_g^\Phi$), the integration rule remains unchanged: the summation still runs over all n_{gp} Gauss points and employs the original FE weights w_g^{FE} .

5.3.1 Hyperreduction (equilibrium)

This observation provides the natural entry point for hyperreduction. From a quadrature (or, more precisely, cubature) perspective, the goal is to construct a reduced integration rule tailored to the reduced integrands $\mathbf{r}_g^\Phi$. This amounts to selecting a subset of Gauss points

$$\mathcal{Z} \subset \{1, \dots, n_{\text{gp}}\},$$

together with associated positive weights w_g^Φ , such that the equilibrium equation can be approximated as

$$\sum_{g \in \mathcal{Z}} \mathbf{r}_g^\Phi(\mathbf{q}; \boldsymbol{\mu}) w_g^\Phi = \mathbf{0}. \quad (11)$$

The construction of $\mathcal{Z}$ and w_g^Φ is therefore a discrete selection problem, in which only a small subset of the original Gauss points is retained.

For a single training state, characterized by $\boldsymbol{\mu}_j$ and by the corresponding ROM coordinates $\mathbf{q}_j$, the projected equilibrium equation can be written as

$$\mathbf{a}_r(\mathbf{q}_j; \boldsymbol{\mu}_j)^T \mathbf{w}^{\text{FE}} = \mathbf{0}, \quad (12)$$

where

$$\mathbf{a}_r(\mathbf{q}_j; \boldsymbol{\mu}_j) = \begin{bmatrix} \mathbf{r}_1^\Phi(\mathbf{q}_j; \boldsymbol{\mu}_j)^T \\ \mathbf{r}_2^\Phi(\mathbf{q}_j; \boldsymbol{\mu}_j)^T \\ \vdots \\ \mathbf{r}_{n_{\text{gp}}}^\Phi(\mathbf{q}_j; \boldsymbol{\mu}_j)^T \end{bmatrix} \in \mathbb{R}^{n_{\text{gp}} \times r}. \quad (13)$$

and

$$\mathbf{w}^{\text{FE}} = \begin{bmatrix} w_1^{\text{FE}} \\ w_2^{\text{FE}} \\ \vdots \\ w_{n_{\text{gp}}}^{\text{FE}} \end{bmatrix} \in \mathbb{R}^{n_{\text{gp}}}, \quad (14)$$

Thus, each column of $\mathbf{a}_r(\mathbf{q}_j; \boldsymbol{\mu}_j)$ represents one scalar reduced-integrand field over the Gauss points of the mesh.

Collecting these matrices for all training states gives the global snapshot matrix of reduced internal-force densities,

$$\mathbf{A}_r = [\mathbf{a}_r(\mathbf{q}_1; \boldsymbol{\mu}_1) \quad \mathbf{a}_r(\mathbf{q}_2; \boldsymbol{\mu}_2) \quad \cdots \quad \mathbf{a}_r(\mathbf{q}_{n_s}; \boldsymbol{\mu}_{n_s})] \in \mathbb{R}^{n_{\text{gp}} \times (rn_s)}. \quad (15)$$

The columns of $\mathbf{A}_r$ span the family of scalar fields that must be integrated accurately by the new cubature rule. Accordingly, the goal is to construct a basis matrix for the column space of $\mathbf{A}_r$ —this is the underlying idea of the Empirical Cubature Method (ECM), see [12]. To this end, a basis matrix

$$\mathbf{U} \in \mathbb{R}^{n_{\text{gp}} \times n_U}, \quad \text{Range}(\mathbf{U}) \approx \text{Range}(\mathbf{A}_r), \quad (16)$$

is computed, typically via a truncated singular value decomposition of $\mathbf{A}_r$, so that $n_U \ll rn_s$ after truncation. A natural choice is to employ a weighted SVD, using the norm induced by the

FE weights. The cubature rule is then constructed so as to integrate exactly the columns of $\mathbf{U}$, rather than those of the full matrix² $\mathbf{A}_r$.

Following the ECM construction [12], the cubature rule is obtained by solving a sparse selection problem. In the present discrete setting, this can be formulated as

$$\begin{aligned} \min_{\mathbf{w}^\Phi} \quad & \|\mathbf{w}^\Phi\|_0 \\ \text{s.t.} \quad & \mathbf{U}^T \mathbf{w}^\Phi = \mathbf{b}^{\text{FE}}, \\ & \mathbf{w}^\Phi \geq 0, \end{aligned} \tag{17}$$

where $\mathbf{w}^\Phi \in \mathbb{R}^{n_{\text{gp}}}$ is the vector of cubature weights, and $\mathbf{b}^{\text{FE}} \in \mathbb{R}^{n_U}$ contains the exact integrals of the columns of $\mathbf{U}$ computed with the FE quadrature rule, i.e.,

$$\mathbf{b}^{\text{FE}} = \mathbf{U}^T \mathbf{w}^{\text{FE}}. \tag{18}$$

Problem (17) seeks the sparsest vector of positive weights such that the selected integration points reproduce exactly the integrals of the basis functions. The support of $\mathbf{w}^\Phi$ defines the reduced set of Gauss points $\mathcal{Z}$, so that the cubature rule is obtained as a discrete subset of the original FE integration points.

5.3.2 Remarks on the ECM formulation

Remark 3 (Well-posedness and inclusion of constants). *In the present setting, the vector of exact integrals $\mathbf{b}^{\text{FE}} = \mathbf{U}^T \mathbf{w}^{\text{FE}}$ is zero (or numerically close to zero), since the snapshots are extracted from converged solutions of the nonlinear equilibrium problem, for which the residual is driven to zero up to the tolerance of the Newton–Raphson procedure. As a consequence, the optimization problem admits the trivial solution $\mathbf{w}^\Phi = \mathbf{0}$.*

To remove this degeneracy [13], we impose the additional normalization constraint that the cubature rule must exactly reproduce the measure of the domain:

$$\mathbf{1}^T \mathbf{w}^\Phi = |\Omega_0|, \tag{19}$$

where $\mathbf{1} \in \mathbb{R}^{n_{\text{gp}}}$ denotes the all-ones vector.

This requirement can be enforced by appending the vector $\mathbf{1}$ to the basis matrix $\mathbf{U}$ (and augmenting $\mathbf{b}^{\text{FE}}$ with the corresponding measure $|\Omega_0|$). Alternatively, to preserve the orthogonality of $\mathbf{U}$ (which is desirable for the efficiency of the ECM algorithm), one may instead add only the component of $\mathbf{1}$ that is orthogonal to the current column space of $\mathbf{U}$. Further details can be found in [13, 12].

Remark 4 (Sparsity and connection with ECSW). *The ℓ_0 formulation in (17) is mainly conceptual and will be useful later. The associated sparsification problem is NP-hard, and ECM does not address it in this strict form. ECM instead relaxes the problem and simply seeks a sparse nonnegative solution of the underdetermined system $\mathbf{U}^T \mathbf{w}^\Phi = \mathbf{b}^{\text{FE}}$. Inspired by the cubature strategy of An et al. [1], ECM is built on a modification of the nonnegative least-squares algorithm of Lawson and Hanson [14]. A distinctive feature of ECM is that the number of selected points is known a priori: it coincides with the number of columns of $\mathbf{U}$, i.e., with the truncated rank of $\mathbf{A}_r$. Thus, once the integrand space has been compressed, the cubature problem reduces to selecting as many integration points as basis functions, together with positive weights.*

²In contrast, related hyperreduction techniques such as the Energy-Conserving Sampling and Weighting (ECSW) method and its variants [10, 11] operate directly on the sampled matrix (or on a subsampled set of its columns), without explicitly constructing a reduced basis for the integrand space. The ECM strategy adopted here separates these two steps: first identifying a low-dimensional space of relevant integrands, and subsequently constructing a cubature rule tailored to that space. In this framework, the approximation error in the integration is introduced at the stage of basis truncation (e.g., via SVD), while the cubature rule is designed to integrate exactly the retained basis functions.

This aspect differs from related hyperreduction techniques such as the Energy-Conserving Sampling and Weighting (ECSW) method³ and its variants [10, 11], also based on the Lawson–Hanson algorithm, which operate directly on the sampled matrix, or on a subset of its columns. In ECM, basis extraction and cubature construction are separated: the integration error is introduced at the stage of SVD truncation of the integrand space, while the retained basis functions are integrated exactly. By contrast, ECSW does not construct an explicit reduced basis for the integrands; therefore, the integration error must be controlled directly at the level of the sampling/weighting procedure, typically through a prescribed tolerance on the residual of the approximation.

5.4 Cubature rule for homogenized PK1 stresses

We now consider the computation of the homogenized PK1 stress, previously written in (6). This quantity is an output functional, not an equilibrium equation. Consequently, its hyperreduction can be addressed in two different ways.

For a single training state $(\mathbf{q}_j, \boldsymbol{\mu}_j)$, define

$$\mathbf{a}_P(\mathbf{q}_j; \boldsymbol{\mu}_j) = \begin{bmatrix} \mathbf{P}_1(\mathbf{q}_j; \boldsymbol{\mu}_j)^T \\ \mathbf{P}_2(\mathbf{q}_j; \boldsymbol{\mu}_j)^T \\ \vdots \\ \mathbf{P}_{n_{\text{gp}}}(\mathbf{q}_j; \boldsymbol{\mu}_j)^T \end{bmatrix} \in \mathbb{R}^{n_{\text{gp}} \times n_\sigma}, \quad (20)$$

where n_σ is the number of PK1 stress components retained in Voigt notation. In the present two-dimensional large-strain setting, $n_\sigma = 4$. Each training state therefore adds n_σ scalar integration constraints.

Collecting all training states gives the stress-output snapshot matrix

$$\mathbf{A}_P = [\mathbf{a}_P(\mathbf{q}_1; \boldsymbol{\mu}_1) \quad \mathbf{a}_P(\mathbf{q}_2; \boldsymbol{\mu}_2) \quad \cdots \quad \mathbf{a}_P(\mathbf{q}_{n_s}; \boldsymbol{\mu}_{n_s})] \in \mathbb{R}^{n_{\text{gp}} \times (n_\sigma n_s)}. \quad (21)$$

This matrix plays for the homogenized stresses the same role that $\mathbf{A}_r$ plays for the reduced internal-force densities.

A first possibility is to use a single cubature rule for both equilibrium and output evaluation. In this case, the matrix used to construct the ECM basis is enriched with the stress-output information contained in $\mathbf{A}_P$, i.e., $\mathbf{U}$ is obtained as the left singular matrix of the concatenation⁴ of $\mathbf{A}_r$ and $\mathbf{A}_P$. The same selected set $\mathcal{Z}$ and weights w_g^Φ are then used both for the reduced residual and for the homogenized PK1 stress:

$$\mathbf{P}^{\text{macro}} \approx \frac{1}{|\Omega_0|} \sum_{g \in \mathcal{Z}} \mathbf{P}_g(\mathbf{q}; \boldsymbol{\mu}) w_g^\Phi. \quad (22)$$

This is the simplest online strategy, since only one set of Gauss points must be stored, evaluated, and differentiated during the Newton iterations.

A second possibility is to construct a cubature rule specifically tailored to the stress-output matrix $\mathbf{A}_P$. This gives a possibly different set of points $\mathcal{Z}_P \subset \{1, \dots, n_{\text{gp}}\}$ and weights w_g^P :

$$\mathbf{P}^{\text{macro}} \approx \frac{1}{|\Omega_0|} \sum_{g \in \mathcal{Z}_P} \mathbf{P}_g(\mathbf{q}; \boldsymbol{\mu}) w_g^P. \quad (23)$$

³ECSW was originally formulated as a mesh (element) selection and weighting strategy rather than as a cubature rule. However, when viewed through the ℓ_0 -based formulation adopted here—actually borrowed from [10]—the two approaches become structurally equivalent when the selected entities are finite elements, since in both cases the goal is to approximate the integrals arising in the weak form using a reduced set of contributions.

⁴Since $\mathbf{A}_P$ and $\mathbf{A}_r$ correspond to different physical quantities (with different units and scales), it is advisable to apply an appropriate normalization prior to the SVD so as to balance their relative influence in the resulting basis.

This route may reduce the number of points required to integrate the output alone, but it is less convenient computationally because two different point sets must be tracked: one for the reduced residual and another one for the homogenized stress. Although the stress output is needed only after convergence of the nonlinear solve, the additional bookkeeping and stress evaluations may still be undesirable. If a separate output rule is used, it is advantageous to maximize its intersection with the residual cubature set $\mathcal{Z}$. For this purpose, one may use the ECM variant proposed by the authors in [6], which accepts an initial candidate set. In the present context, $\mathcal{Z}$ is taken as such initial set; the algorithm first attempts to satisfy the stress integration constraints using these points and enlarges the set only if necessary, thereby maximizing the overlap between both cubature rules.

5.5 Numerical assessment

We now turn our attention to the numerical assessment of the performance of the ECM in the case discussed previously, in which the nodal displacement field requires $r = 52$ modes to achieve a relative accuracy of 10^{-6} . We adopt here the strategy in which a single cubature rule is used for both equilibrium and stress integration, cf. Eq. 22. The underlying finite element discretization involves a total of $n_{\text{gp}} = n_{\text{el}} n_{\text{gp}}^e = 5260$ Gauss integration points. For each instantiation of the input parameter, the corresponding contribution to $[\mathbf{A}_r, \mathbf{A}_p]$ consists of $r + n_\sigma$ columns, accounting for the reduced equilibrium equations together with the stress-output contributions. Over the full training set of $n_s = 4852$ states, this leads to a total of $(r + n_\sigma)n_s = (52 + 4) \times 4852 = 271712$ columns (recall that $n_\sigma = 4$ is the number of PK1 stress components retained in Voigt notation). It follows that the matrix whose column space must be approximated has size 5260×271712 , that is, it is markedly wide, with roughly fifty times more columns than rows.

This situation departs fundamentally from the standard paradigm of reduced-order modeling, in which snapshot matrices are typically tall and thin and therefore amenable to low-rank approximation. Here, by contrast, a very large number of scalar integrand fields must be represented simultaneously, each corresponding to an independent reduced equilibrium or stress constraint. The size of the matrix further reflects this fact. Assuming double-precision storage, its memory footprint exceeds 11 GB, which makes its explicit assembly impractical. To address this difficulty, the matrix is processed sequentially using the randomized SVD strategy proposed in [12]. In this approach, blocks of size $n_{\text{gp}} \times (r + n_\sigma)$ are compressed independently with a prescribed tolerance of 10^{-4} , and the resulting subspaces are merged progressively so as to approximate the global column space. This procedure avoids the explicit assembly of the full matrix while preserving control over the approximation error. This procedure yields 1339 modes. Enforcing exact integration of constant fields requires the inclusion of one additional mode, leading to a final total of 1340 basis functions. This implies that the number of selected Gauss points is also 1340. This corresponds to approximately 25.5% of the original quadrature rule—that is, the number of retained integration points remains of the same order of magnitude as the original one, and the resulting computational savings are therefore limited.

The spatial distribution of the selected integration points is shown in Fig. 7. A total of 671 elements are retained. As can be observed, the selected regions are distributed across the entire domain, rather than being concentrated in localized areas. This behavior reflects the complexity of the underlying integrand fields, which are influenced by nonlinear deformation mechanisms and cannot be accurately represented using a small number of localized sampling points.

Taken together, these observations lead to a clear conclusion. Despite the low-dimensional parametrization of the problem, the associated integrand space exhibits a high intrinsic complexity, both in terms of dimensionality and spatial distribution. As a result, the DECM requires a large number of modes and, consequently, a large number of integration points. This directly limits the effectiveness of any fixed cubature rule.

Therefore, in the present setting, a standard hyperreduced-order model based on a fixed ECM

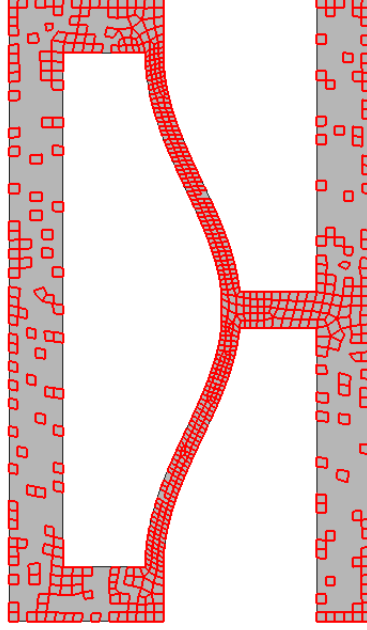


Figure 7: Elements containing the selected Gauss points in $\mathcal{Z}$ (in red). A total of 671 elements are retained, corresponding to 1340 integration points.

rule is not a viable approach. The lack of strong compressibility of the integrand space prevents the construction of a sufficiently sparse integration rule, and thus precludes the possibility of achieving significant computational gains within the classical HROM framework.

The construction of the ECM integration basis is implemented in blocks. First, the reduced displacement-gradient operator is assembled in `OFFLINE_STAGE_nonECM_genFASTd.m`. For periodic boundary conditions, the relevant operator is built as `BstRED_1 = OPERFE.Bst*OTHER.output.DISP.CONDITIONS.A*Bas`, which corresponds to the product between the Gauss-point gradient operator, the periodic lifting matrix, and the displacement basis.

Although the main text presents the homogenized PK1 stress through volume averaging, the current implementation evaluates the corresponding output through boundary reactions. The reaction-output modes are computed in `DetermineModeReactionsOUTPUT_glo.m`, which calls `DetermineModeHomogViaREACT.m`. The resulting reaction operator is denoted in the code by `BstRED_r`. When reaction outputs are included in the ECM construction, this block may be normalized by its Frobenius norm and rescaled with the Frobenius norm of the equilibrium block, in order to balance both contributions in the SVD.

The flags controlling the construction are `DATAoffline.ECMforEquilibrium` and `DATAoffline.ECMforReactionForces`. If only equilibrium is considered, the integrand matrix is assembled from the equilibrium block. If reaction forces are also considered, the reaction-output block is appended to the equilibrium block before constructing the ECM basis. If equilibrium is disabled and reaction forces are enabled, the ECM basis is constructed only from the reaction-output block.

The matrix of reduced internal-force densities is assembled in `OFFLINE_STAGE_nonECM_genFASTd.m` through calls to `BasisF_from_BasisStress_PK1.m`. This routine receives the reduced gradient block, or the concatenation of the equilibrium and reaction-output blocks, together with the PK1 stress snapshots, and returns the matrix used as input for the SVD that defines the ECM basis.

The basis matrix $\mathbf{U}$ is computed in `QbasisMatrixIntegrand_givenSNAPr.m`. The SVD can be performed in the norm induced by the FE quadrature weights. When the full matrix does not fit in memory, the partitioned implementation `Qmatrix_LOC_forECM_cell.m` is used. This version processes the integrand matrix by blocks and employs the SRSVD strategy introduced in [12].

The same routines also implement the enrichment required to integrate the constant function. The component of the volume vector not represented by the current SVD basis is computed, orthonormalized if necessary, and appended to $\mathbf{U}$. This is the implementation counterpart of the additional volume constraint discussed in the main text.

Once $\mathbf{U}$ has been constructed, the discrete ECM problem is launched from `DiscreteECM_givenAmat.m`. The actual greedy cubature algorithm is implemented in `DiscreteEmpiricalCubatureMethod.m`, which returns the selected Gauss points and their positive cubature weights.

5.6 A posteriori accuracy of the standard HROM

We now assess the accuracy of the standard HROM obtained from the linear basis Φ . The comparison is performed on the training trajectories, using the FE solution as reference. We report the relative Frobenius displacement error e_d , together with two indicators for the homogenized PK1 stresses: the relative Frobenius error e_P and the relative maximum error $e_P^{\max}$. The values in Table 1 correspond to averages over the two loading trajectories in the macroscopic parameter space.

Table 1: A posteriori errors for the standard HROM. The displacement tolerance controls the number of retained displacement modes, while the internal-force SVD tolerance is kept fixed at 10^{-4} in the ECM cases.

Case	$\varepsilon_d^{\text{SVD}}$	r	n_{ECM}	e_d	e_P
Full Gauss rule	10^{-6}	52	5260	2.52×10^{-6}	5.05×10^{-5}
ECM	10^{-6}	52	1340	1.85×10^{-5}	7.13×10^{-5}
ECM	10^{-5}	37	1173	4.22×10^{-5}	5.93×10^{-4}
ECM	10^{-4}	25	1011	4.99×10^{-4}	9.28×10^{-3}

The first two rows isolate the effect of hyperreduction. With $r = 52$, the use of ECM increases the displacement error from 2.52×10^{-6} to 1.85×10^{-5} , and the homogenized-stress error from 5.05×10^{-5} to 7.13×10^{-5} . Thus, for this tolerance level, the ECM contribution to the error is visible but remains small.

The more significant trend appears when the displacement basis is truncated more aggressively. Reducing the SVD tolerance from 10^{-6} to 10^{-5} decreases the number of modes from 52 to 37, but the number of ECM points only decreases from 1340 to 1173. With a tolerance of 10^{-4} , the number of modes is further reduced to 25, while the ECM rule still requires 1011 points. This weak decrease of n_{ECM} with r is consistent with the observation that the dimension of the space spanned by the reduced internal-force densities may scale much faster than the displacement basis itself, as discussed in [6].

More importantly, the homogenized stresses are much more sensitive than the global displacement error. With 25 modes, the average displacement error remains below 5×10^{-4} , but the average PK1 stress error rises to approximately 9.3×10^{-3} , with componentwise maximum errors exceeding 3% in some parts of the trajectory. This shows that the Frobenius norm of the displacement snapshots is not, by itself, a reliable indicator of output accuracy: local kinematic errors that are small in a global norm may have a significant impact on stress quantities.

This effect is illustrated in Figures 8 and 9. The response combines a relatively smooth tensile branch with a compression branch dominated by curvature changes and buckling-like deformation of the curved ligaments. These localized features have limited weight in a global displacement SVD, but they are essential for reproducing the homogenized stress. Consequently, modes that may appear secondary from a purely Frobenius-norm viewpoint can be mechanically relevant.

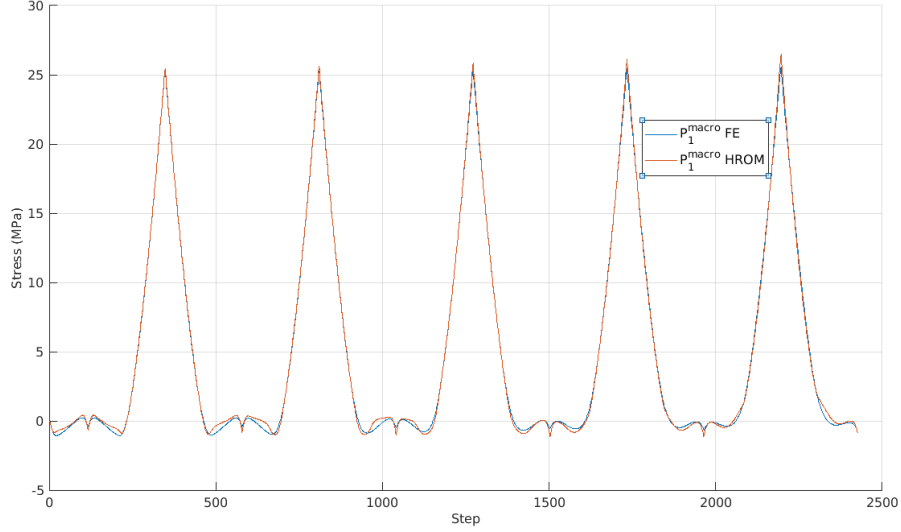


Figure 8: Comparison between FE and HROM homogenized PK1 stress histories using 25 displacement modes. The largest discrepancies occur in the compression-dominated part of the trajectory, where the response is governed by localized changes of curvature and buckling-like deformation of the curved ligaments.

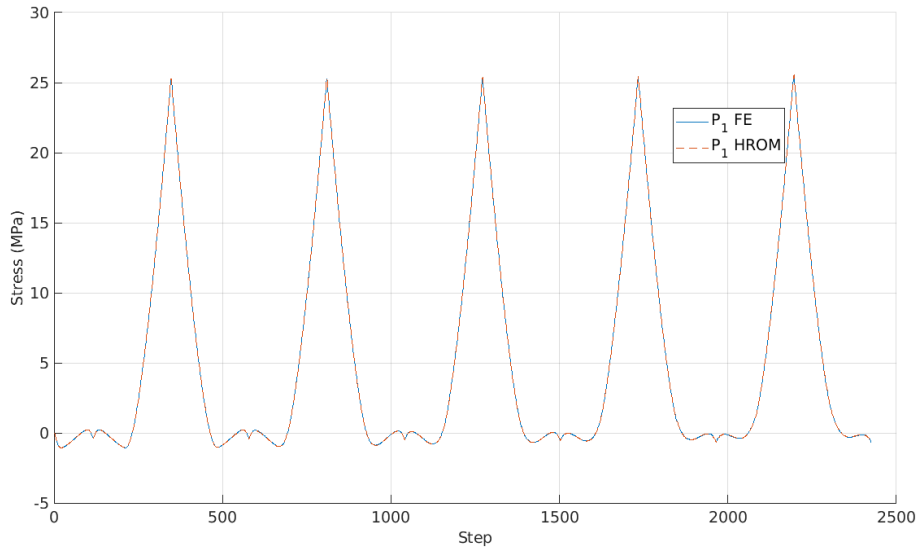


Figure 9: Comparison between FE and HROM homogenized PK1 stress histories using 37 displacement modes. Increasing the number of modes substantially improves the stress prediction, confirming that stress accuracy requires retaining displacement modes associated with localized mechanisms.

The consistency check is performed in `CHECK_consistency_non_gen3Rg.m`. The same routine is used for standard and manifold HROMs; the standard HROM is recovered as the particular case in which the nonlinear map is the identity (`tauFUN_identity`). Homogenized PK1 stress errors are computed in `CompareHomogenizedPK1Stresses.m`. The cases reported in Table 1 correspond to `NameInputFile = 'Data_M3std'`, `'Data_M3stdNH'`, `'Data_M3std2'`, and `'Data_M3std3'`.

The figures shown in Section 5.6 are generated from the MATLAB files `PK1_ECM.case4.fig` and `PK1_1comp.37.fig`.

6 Nonlinear-manifold reduced-order model

The numerical evidence reported in Section 5.6 shows that the linear approximation introduced in Eq. 7 does not provide a sufficiently compact description of the present problem. Although the macroscopic input space is two-dimensional, the microscopic response involves localized changes of curvature and buckling-like deformation of the curved ligaments, especially along the compression branch. These features are poorly represented by a global linear subspace: modes that carry little energy in the Frobenius norm may still be essential for the accurate prediction of homogenized stresses.

This behavior is consistent with the Kolmogorov-width limitation of linear projection-based reduced-order models. A solution set may be parameterized by a small number of variables and nevertheless require a relatively large linear space for accurate approximation. In the present case, this limitation is even more visible at the hyperreduction level. The number of ECM points decreases only weakly with the number of retained displacement modes, indicating that the space of reduced internal-force and stress-output integrands has a complexity that is not controlled directly by the dimension of the displacement subspace. For nonlinear problems, this integrand space may grow rapidly with the number of reduced coordinates, since the residual depends nonlinearly on the reduced state and its projection introduces products between tangent directions and state-dependent stress fields.

These observations motivate the replacement of the global linear subspace by a nonlinear approximation manifold. The objective is not to abandon the structure of the finite element equilibrium problem, but to modify the kinematic approximation used to represent the independent displacement vector. Following the terminology commonly used in nonlinear projection-based model reduction, we introduce a decoder

$$\mathcal{D} : \mathbb{R}^{r_{\mathcal{D}}} \rightarrow \mathbb{R}^{n_L},$$

and approximate the independent displacement vector as

$$\mathbf{d}_L \approx \mathcal{D}(\mathbf{q}_M), \quad (24)$$

where $\mathbf{q}_M \in \mathbb{R}^{r_{\mathcal{D}}}$ denotes the nonlinear-manifold coordinates⁵. The dimension $r_{\mathcal{D}}$ is expected to be much smaller than the dimension r required by the linear ROM for comparable accuracy. Notice that the linear approximation in Eq. 7 is recovered as the particular case $\mathcal{D}(\mathbf{q}_M) = \Phi \mathbf{q}_M$, up to the notational identification of $\mathbf{q}_M$ with the standard reduced coordinate vector $\mathbf{q}$. Thus, the nonlinear-manifold approximation is not a different mechanical model, but a generalization of the trial space used in the projection.

For offline construction and post-processing it is also convenient to introduce the inverse operation, or encoder,

$$\mathcal{E} : \mathbb{R}^{n_L} \rightarrow \mathbb{R}^{r_{\mathcal{D}}}, \quad \mathbf{q}_M \approx \mathcal{E}(\mathbf{d}_L).$$

This map assigns manifold coordinates to a given displacement snapshot. In the present work, the encoder is used only as an offline device to parameterize the available finite element snapshots and to organize the quantities required for hyperreduction. The formulation below does not assume that the nonlinear manifold is obtained from an autoencoder; the terms encoder and decoder are used only to emphasize the analogy with latent-variable representations.

6.1 Tangent-space projection

The nonlinear approximation in Eq. 24 induces a state-dependent space of admissible virtual displacements. Taking the first variation gives

$$\delta \mathbf{d}_L = \frac{\partial \mathcal{D}}{\partial \mathbf{q}_M}(\mathbf{q}_M) \delta \mathbf{q}_M. \quad (25)$$

⁵The meaning of the subscript M will become apparent later. At this stage, it only identifies the coordinates used to parameterize the nonlinear reduced manifold

We denote the Jacobian of the decoder by

$$\Phi_{\mathcal{D}}(\mathbf{q}_M) := \frac{\partial \mathcal{D}}{\partial \mathbf{q}_M}(\mathbf{q}_M) \in \mathbb{R}^{n_L \times r_{\mathcal{D}}}. \quad (26)$$

The columns of $\Phi_{\mathcal{D}}(\mathbf{q}_M)$ span the tangent space to the nonlinear approximation manifold at the state $\mathcal{D}(\mathbf{q}_M)$. This is the main difference with the standard ROM: in Eq. 7, the test space is the fixed linear space spanned by Φ , whereas in the present formulation the admissible virtual displacements depend on the current manifold coordinate $\mathbf{q}_M$.

The constrained finite element equilibrium equation remains that given in Eq. 11, with the only modification being the choice of test space. Projection onto the tangent space of the nonlinear manifold yields

$$\sum_{g=1}^{n_{\text{gp}}} \mathbf{r}_g^{\mathcal{D}}(\mathbf{q}_M; \boldsymbol{\mu}) w_g^{\text{FE}} = \mathbf{0}, \quad (27)$$

where

$$\mathbf{r}_g^{\mathcal{D}}(\mathbf{q}_M; \boldsymbol{\mu}) := \Phi_{\mathcal{D}}(\mathbf{q}_M)^T \mathbf{r}_g^{\text{FE}}(\mathbf{q}_M; \boldsymbol{\mu}) \in \mathbb{R}^{r_{\mathcal{D}}}. \quad (28)$$

This formulation defines a system of $r_{\mathcal{D}}$ nonlinear equations for the manifold coordinates $\mathbf{q}_M$. At this stage, the integration rule remains unchanged: all n_{gp} Gauss points are retained together with the original finite element weights w_g^{FE} .

6.2 Fixed-weight hyperreduction on the nonlinear manifold

The expression (28) highlights the key structural difference with the linear setting. The scalar integrands are now obtained by projection onto the state-dependent tangent basis $\Phi_{\mathcal{D}}(\mathbf{q}_M)$, rather than onto a fixed subspace. Nevertheless, before introducing adaptive weights, it is useful to formulate the direct nonlinear-manifold counterpart of the standard ECM-based HROM.

In this subsection we retain the same hyperreduction paradigm adopted in Section 5.6: the cubature weights are fixed, i.e., independent of the current reduced state. Thus, the full Gauss-point equilibrium equation (27) is replaced by

$$\sum_{g \in \mathcal{Z}_{\mathcal{D}}} \mathbf{r}_g^{\mathcal{D}}(\mathbf{q}_M; \boldsymbol{\mu}) w_g^{\mathcal{D}} = \mathbf{0}, \quad (29)$$

where $\mathcal{Z}_{\mathcal{D}} \subset \{1, \dots, n_{\text{gp}}\}$ is the selected set of Gauss points and $w_g^{\mathcal{D}} > 0$ are the associated cubature weights. The superscript $\mathcal{D}$ emphasizes that the weights are attached to the decoder-based manifold formulation.

As in the standard HROM, we consider here the case in which the same cubature rule is used for both the governing equilibrium equations and the output of interest. Therefore, the homogenized PK1 stress is approximated as

$$\mathbf{P}^{\text{macro}} \approx \frac{1}{|\Omega_0|} \sum_{g \in \mathcal{Z}_{\mathcal{D}}} \mathbf{P}_g(\mathbf{q}_M; \boldsymbol{\mu}) w_g^{\mathcal{D}}. \quad (30)$$

The notation $\mathbf{P}_g(\mathbf{q}_M; \boldsymbol{\mu})$ means that the microscopic stress is evaluated from the decoder-reconstructed displacement $\mathcal{AD}(\mathbf{q}_M) + \bar{\mathbf{U}}\boldsymbol{\mu}$.

The training matrix associated with the equilibrium integrands is modified accordingly. For a single training state $(\mathbf{q}_{Mj}, \boldsymbol{\mu}_j)$, define

$$\mathbf{a}_{\mathcal{D}}(\mathbf{q}_{Mj}; \boldsymbol{\mu}_j) = \begin{bmatrix} \mathbf{r}_1^{\mathcal{D}}(\mathbf{q}_{Mj}; \boldsymbol{\mu}_j)^T \\ \mathbf{r}_2^{\mathcal{D}}(\mathbf{q}_{Mj}; \boldsymbol{\mu}_j)^T \\ \vdots \\ \mathbf{r}_{n_{\text{gp}}}^{\mathcal{D}}(\mathbf{q}_{Mj}; \boldsymbol{\mu}_j)^T \end{bmatrix} \in \mathbb{R}^{n_{\text{gp}} \times r_{\mathcal{D}}}. \quad (31)$$

Collecting all training states gives

$$\mathbf{A}_{\mathcal{D}} = [\mathbf{a}_{\mathcal{D}}(\mathbf{q}_{M1}; \boldsymbol{\mu}_1) \quad \mathbf{a}_{\mathcal{D}}(\mathbf{q}_{M2}; \boldsymbol{\mu}_2) \quad \cdots \quad \mathbf{a}_{\mathcal{D}}(\mathbf{q}_{Mn_s}; \boldsymbol{\mu}_{n_s})] \in \mathbb{R}^{n_{\text{gp}} \times (r_{\mathcal{D}} n_s)}. \quad (32)$$

This matrix is the nonlinear-manifold counterpart of Eq. ???. The difference is purely structural but important: in the standard HROM, each state contributes r scalar integrand fields, one for each column of the fixed basis $\boldsymbol{\Phi}$; in the manifold formulation, each state contributes only $r_{\mathcal{D}}$ fields, associated with the tangent basis $\boldsymbol{\Phi}_{\mathcal{D}}(\mathbf{q}_{Mj})$. In favorable cases, $r_{\mathcal{D}}$ can be chosen as small as the dimension n_{μ} of the macroscopic input space.

The stress-output matrix $\mathbf{A}_P$ keeps the same structure as in Eq. 15. What changes is the state at which the Gauss-point stresses are evaluated: they are now computed from the decoder-reconstructed displacement on the nonlinear manifold. If the same cubature rule is required to approximate both equilibrium and output quantities, the matrix used to construct the ECM basis is obtained from the concatenation

$$[\mathbf{A}_{\mathcal{D}} \quad \mathbf{A}_P], \quad (33)$$

with the same caveat as before regarding the relative scaling of equilibrium and stress-output blocks.

Once the corresponding left singular basis has been computed, the cubature problem is formally identical to the standard ECM problem 17. With a slight abuse of notation, we again denote this basis by $\mathbf{U}$ and its exact integral vector by $\mathbf{b} = \mathbf{U}^T \mathbf{w}^{\text{FE}}$, now omitting the superscript “FE” for compactness. The fixed manifold-HROM cubature rule is obtained from

$$\begin{aligned} \min_{\mathbf{w}^{\mathcal{D}}} \quad & \|\mathbf{w}^{\mathcal{D}}\|_0 \\ \text{s.t.} \quad & \mathbf{U}^T \mathbf{w}^{\mathcal{D}} = \mathbf{b}, \\ & \mathbf{w}^{\mathcal{D}} \geq 0. \end{aligned} \quad (34)$$

The support of $\mathbf{w}^{\mathcal{D}}$ defines $\mathcal{Z}_{\mathcal{D}}$. Thus, after the integrand matrix has been assembled, the construction of a fixed-weight manifold HROM follows the same ECM machinery as in the linear case.

6.2.1 Construction of the manifold ECM snapshots

A subtle point concerns the construction of the snapshot matrices entering Eq. 33. In the standard HROM this step is essentially implicit: the FE snapshots are projected onto the fixed linear basis and the required Gauss-point quantities are evaluated at the corresponding reduced states. In the nonlinear-manifold setting, the same operation requires specifying how the training states are placed on the manifold.

One possibility is to solve the nonlinear-manifold ROM using the full Gauss rule and then use the resulting states to assemble $\mathbf{A}_{\mathcal{D}}$ and $\mathbf{A}_P$. This is conceptually direct, but it requires an additional full-integration reduced solve for each training configuration.

The alternative adopted here is to use the available FE snapshots. Given a finite element displacement snapshot $\mathbf{d}_L(\boldsymbol{\mu}_j)$, the encoder provides the associated manifold coordinate,

$$\mathbf{q}_{Mj} = \mathcal{E}(\mathbf{d}_L(\boldsymbol{\mu}_j)).$$

The state used for stress and internal-force evaluation is then not the raw FE snapshot itself, but its decoder reconstruction,

$$\mathbf{d}_{Lj}^{\mathcal{D}} := \mathcal{D}(\mathcal{E}(\mathbf{d}_L(\boldsymbol{\mu}_j))). \quad (35)$$

The Gauss-point strains, stresses, internal-force contributions, and tangent projections entering $\mathbf{A}_{\mathcal{D}}$ and $\mathbf{A}_P$ are evaluated from $\mathbf{A} \mathbf{d}_{Lj}^{\mathcal{D}} + \bar{\mathbf{U}} \boldsymbol{\mu}_j$. This distinction is important: the ECM training matrix must be consistent with the approximation manifold on which the online HROM will operate, rather than with displacement fields that do not belong exactly to that manifold.

Remark 5 (Input-driven manifolds in static homogenization). *There is a special situation, relevant for the present homogenization example, in which the construction simplifies further. The macroscopic input is a displacement-like quantity imposed through the boundary conditions, the problem is static, and the material response is hyperelastic, so there is no dependence on loading history or inertial effects. Under these assumptions, the microscopic equilibrium solution is uniquely parameterized by the prescribed macroscopic deformation. Consequently, one may choose the manifold coordinates to coincide with the macroscopic input variables,*

$$\mathbf{q}_M \equiv \boldsymbol{\mu}, \quad r_{\mathcal{D}} = n_{\mu}.$$

This is a very specific setting and should not be interpreted as a general property of nonlinear manifold ROMs. Its consequence, however, is important: if the goal is only to approximate the output map $\boldsymbol{\mu} \mapsto \mathbf{P}^{\text{macro}}(\boldsymbol{\mu})$, then solving the reduced equilibrium equation is no longer necessary. In that case, the cubature rule does not need to reproduce the manifold-projected equilibrium integrands in $\mathbf{A}_{\mathcal{D}}$; it only needs to integrate accurately the stress-output fields. The matrix used to construct $\mathbf{U}$ is then simply $\mathbf{A}_P$. Since $\mathbf{A}_P$ has only $n_{\sigma}n_s$ columns, instead of $(r_{\mathcal{D}} + n_{\sigma})n_s$ columns when both equilibrium and output constraints are retained, the rank of the ECM basis may be significantly smaller, and the resulting cubature rule may require fewer integration points.

6.3 Identification of the nonlinear manifold

We now particularize the fixed-weight nonlinear-manifold HROM introduced above to the meta-material cell. The first step is the identification of the nonlinear approximation manifold, namely the construction of the latent coordinates $\mathbf{q}_M$ and of the decoder $\mathcal{D}$. This point was deliberately left open in the abstract formulation, since the hyperreduction strategy does not depend on a unique choice of manifold-learning procedure.

The starting point is the same linear basis $\boldsymbol{\Phi} \in \mathbb{R}^{n_L \times r}$ used in the standard HROM. Rather than using this basis as the final trial space, we use it as an intermediate coordinate system in which a nonlinear approximation manifold is constructed. The manifold approximation is written as

$$\mathbf{d}_L \approx \mathcal{D}(\mathbf{q}_M) := \boldsymbol{\Phi}_M \mathbf{A}_M \mathbf{q}_M + \boldsymbol{\Phi}_S \mathcal{N}_S(\mathbf{q}_M). \quad (36)$$

where

$$\boldsymbol{\Phi}_M^T \boldsymbol{\Phi}_S = \mathbf{0} \quad (37)$$

Here $\boldsymbol{\Phi}_M \in \mathbb{R}^{n_L \times r_{\mathcal{D}}}$ denotes the master basis, $\boldsymbol{\Phi}_S \in \mathbb{R}^{n_L \times (r - r_{\mathcal{D}})}$ is the complementary slave basis, and

$$\mathcal{N}_S : \mathbb{R}^{r_{\mathcal{D}}} \rightarrow \mathbb{R}^{r - r_{\mathcal{D}}}$$

is the nonlinear map that reconstructs the slave amplitudes from the master coordinates.

We next describe how $\boldsymbol{\Phi}_M$, $\mathbf{A}_M$, and $\mathbf{q}_M$ are determined. Let $\mathbf{D}_L \in \mathbb{R}^{n_L \times n_s}$ be the displacement snapshot matrix introduced in Eq. 8. The corresponding matrix of POD coefficients is

$$\mathbf{Q}_{ROM} = \boldsymbol{\Phi}^T \mathbf{D}_L = [\mathbf{q}_1 \quad \mathbf{q}_2 \quad \cdots \quad \mathbf{q}_{n_s}] \in \mathbb{R}^{r \times n_s}.$$

We also define

$$\mathbf{M}_{\mu} = [\boldsymbol{\mu}_1 \quad \boldsymbol{\mu}_2 \quad \cdots \quad \boldsymbol{\mu}_{n_s}] \in \mathbb{R}^{n_{\mu} \times n_s},$$

where $\mathbf{M}_{\mu}$ collects the macroscopic input coordinates associated with the training snapshots. In the present case, $n_{\mu} = 2$ and $\boldsymbol{\mu} = (G_x^{\text{macro}}, G_{xy}^{\text{macro}})^T$. We seek a linear map

$$\mathbf{T}_M \in \mathbb{R}^{r_{\mathcal{D}} \times r}, \quad r_{\mathcal{D}} = n_{\mu} = 2,$$

such that the reduced coordinates are mapped onto the macroscopic input coordinates in the least-squares sense:

$$\mathbf{T}_M = \arg \min_{\mathbf{T}_M} \left\| \widetilde{\mathbf{T}_M} \mathbf{Q}_{ROM} - \mathbf{M}_\mu \right\|_F^2. \quad (38)$$

If the residual in Eq. 38 is negligible, the master coordinates can be identified with the macroscopic input variables,

$$\mathbf{q}_M = \mathbf{T}_M \mathbf{q} \approx \boldsymbol{\mu}. \quad (39)$$

This is the input-driven situation anticipated in Remark 5.

The master displacement subspace associated with $\mathbf{T}_M$ is first obtained in non-orthonormal form as

$$\boldsymbol{\Phi}_c = \boldsymbol{\Phi} \mathbf{T}_M^T \in \mathbb{R}^{n_L \times r_D}. \quad (40)$$

Indeed, since $\mathbf{q}_M = \mathbf{T}_M \mathbf{q}$ and $\mathbf{q} = \boldsymbol{\Phi}^T \mathbf{d}_L$, one has

$$\mathbf{q}_M = \mathbf{T}_M \boldsymbol{\Phi}^T \mathbf{d}_L = \boldsymbol{\Phi}_c^T \mathbf{d}_L.$$

Thus, the columns of $\boldsymbol{\Phi}_c$ define the linear functionals that extract the master coordinates from a displacement field.

Since $\boldsymbol{\Phi}_c$ is not necessarily orthonormal, we compute the thin SVD

$$\boldsymbol{\Phi}_c = \boldsymbol{\Phi}_M \boldsymbol{\Sigma}_M \mathbf{V}_M^T,$$

and define $\boldsymbol{\Phi}_M$ as the orthonormal master basis. If $\mathbf{d}_{LM} = \boldsymbol{\Phi}_M \mathbf{a}_M$ denotes the master component of the displacement, then

$$\mathbf{q}_M = \boldsymbol{\Phi}_c^T \mathbf{d}_{LM} = \mathbf{V}_M \boldsymbol{\Sigma}_M \mathbf{a}_M.$$

Therefore

$$\mathbf{a}_M = \boldsymbol{\Sigma}_M^{-1} \mathbf{V}_M^T \mathbf{q}_M,$$

and the matrix appearing in the decoder is

$$\mathbf{A}_M = \boldsymbol{\Sigma}_M^{-1} \mathbf{V}_M^T. \quad (41)$$

Consequently, the master contribution in Eq. 36 is written as $\boldsymbol{\Phi}_M \mathbf{A}_M \mathbf{q}_M$.

The slave basis is then constructed as the orthogonal complement of $\boldsymbol{\Phi}_M$ inside $\text{Col}(\boldsymbol{\Phi})$ (the column space of $\boldsymbol{\Phi}$). More precisely, the component of $\boldsymbol{\Phi}$ already represented by $\boldsymbol{\Phi}_M$ is removed and the remaining subspace is orthonormalized. This yields the decomposition stated in Eq. 37, up to the truncation tolerance used in the SVD. The slave amplitudes associated with the training snapshots are then computed as

$$\mathbf{q}_{Sj} = \boldsymbol{\Phi}_S^T \mathbf{d}_L(\boldsymbol{\mu}_j),$$

and the nonlinear closure relation

$$\mathbf{q}_S \approx \mathcal{N}_S(\mathbf{q}_M) \quad (42)$$

is learned from the training pairs $(\mathbf{q}_{Mj}, \mathbf{q}_{Sj})$. This last regression step is discussed later in Section 6.4. The construction is inspired by nonlinear projection-based ROMs on approximation manifolds developed by Farhat and co-workers [3, 4, 2], but differs in a key aspect. In the primary–secondary strategy proposed in these works, the primary (master) variables are typically chosen as the leading POD amplitudes, which amounts to taking $\mathbf{T}_M$ as a Boolean selection matrix of the form

$$\mathbf{T}_M^{\text{POD}} = [\mathbf{I}_{r_D} \quad \mathbf{0}] \in \mathbb{R}^{r_D \times r},$$

(here $\mathbf{I}_{r_D} \in \mathbb{R}^{r_D \times r_D}$ denotes the identity matrix of size r_D) so that $\mathbf{q}_M$ consists of the first r_D coefficients of $\mathbf{q}$. While this approach is well suited to applications in which no clear low-dimensional parametrization is available, it does not, in general, provide a direct mechanism to guarantee minimality of the latent dimension.

6.3.1 Encoder

We now define the encoder associated with the nonlinear manifold. In general, for nonlinear approximation manifolds, the reduced coordinates associated with a displacement snapshot are obtained by solving a nonlinear least-squares problem.

$$\mathcal{E}(\mathbf{d}_L) = \arg \min_{\widetilde{\mathbf{q}}_M \in \mathbb{R}^{r_D}} \frac{1}{2} \|\mathbf{d}_L - \mathcal{D}(\widetilde{\mathbf{q}}_M)\|_2^2. \quad (43)$$

This is the standard construction used in nonlinear projection-based ROMs on approximation manifolds; see, for instance, the Gauss–Newton encoder procedure described in [2]. The stationarity condition associated with Eq. 43 reads

$$\mathbf{J}_{\mathcal{D}}(\mathbf{q}_M)^T [\mathcal{D}(\mathbf{q}_M) - \mathbf{d}_L] = \mathbf{0}, \quad \mathbf{J}_{\mathcal{D}}(\mathbf{q}_M) := \frac{\partial \mathcal{D}}{\partial \mathbf{q}_M}(\mathbf{q}_M). \quad (44)$$

For the decoder used here,

$$\mathcal{D}(\mathbf{q}_M) = \Phi_M \mathbf{A}_M \mathbf{q}_M + \Phi_S \mathcal{N}_S(\mathbf{q}_M),$$

the tangent matrix is

$$\mathbf{J}_{\mathcal{D}}(\mathbf{q}_M) = \Phi_M \mathbf{A}_M + \Phi_S \frac{\partial \mathcal{N}_S}{\partial \mathbf{q}_M}(\mathbf{q}_M). \quad (45)$$

Substituting Eq. 45 into Eq. 44 gives

$$\left(\mathbf{A}_M^T \Phi_M^T + \frac{\partial \mathcal{N}_S}{\partial \mathbf{q}_M}(\mathbf{q}_M)^T \Phi_S^T \right) [\Phi_M \mathbf{A}_M \mathbf{q}_M + \Phi_S \mathcal{N}_S(\mathbf{q}_M) - \mathbf{d}_L] = \mathbf{0}. \quad (46)$$

Using the orthogonality of the master and slave bases, this expression reduces to

$$\mathbf{A}_M^T (\mathbf{A}_M \mathbf{q}_M - \Phi_M^T \mathbf{d}_L) + \frac{\partial \mathcal{N}_S}{\partial \mathbf{q}_M}(\mathbf{q}_M)^T (\mathcal{N}_S(\mathbf{q}_M) - \Phi_S^T \mathbf{d}_L) = \mathbf{0}. \quad (47)$$

Equation 47 shows explicitly why the exact encoder is nonlinear: the slave contribution enters through the derivative of the nonlinear map. To interpret this term, let the manifold approximation be written as

$$\mathbf{d}_L = \Phi_M \mathbf{A}_M \mathbf{q}_M + \Phi_S \mathcal{N}_S(\mathbf{q}_M) + \mathbf{e}_{\mathcal{D}}, \quad (48)$$

where $\mathbf{e}_{\mathcal{D}} \in \mathbb{R}^{n_L}$ denotes the manifold approximation error. Premultiplication by Φ_S^T gives

$$\Phi_S^T \mathbf{d}_L = \mathcal{N}_S(\mathbf{q}_M) + \Phi_S^T \mathbf{e}_{\mathcal{D}},$$

and therefore

$$\mathcal{N}_S(\mathbf{q}_M) - \Phi_S^T \mathbf{d}_L = -\Phi_S^T \mathbf{e}_{\mathcal{D}}.$$

Hence, the second term in Eq. 47 is directly controlled by the manifold approximation error. When the manifold representation is sufficiently accurate, this contribution becomes small and the stationarity condition reduces approximately to

$$\mathbf{A}_M^T (\mathbf{A}_M \mathbf{q}_M - \Phi_M^T \mathbf{d}_L) = \mathbf{0},$$

which yields

$$\mathbf{q}_M = \mathbf{A}_M^{-1} \Phi_M^T \mathbf{d}_L. \quad (49)$$

Empirically, this simplified encoder has been found sufficiently accurate for the present application.

The encoder is called in `OFFLINE_STAGE.nonECM.2var.m` through `Encoder1MasterMode.Gen.m`. The implementation computes the projection of the displacement snapshots onto the master basis, rescales the result with `DATA.evaluateTAU.and.DER.Am`, and obtains the master coordinates. The nonlinear decoder is implemented through the ROM-coordinate vector $\boldsymbol{\tau}(\mathbf{q}_M)$, so that

$$\mathcal{D}(\mathbf{q}_M) = \Phi \boldsymbol{\tau}(\mathbf{q}_M).$$

With the present master–slave convention,

$$\boldsymbol{\tau}(\mathbf{q}_M) = \begin{bmatrix} \mathbf{A}_M \mathbf{q}_M \\ \mathcal{N}_S(\mathbf{q}_M) \end{bmatrix}.$$

The function `EvalTauSlaveRBF_Online.m` evaluates $\boldsymbol{\tau}(\mathbf{q}_M)$ and, when requested, its derivatives. Its Jacobian is

$$\frac{\partial \boldsymbol{\tau}}{\partial \mathbf{q}_M} = \begin{bmatrix} \mathbf{A}_M \\ \frac{\partial \mathcal{N}_S}{\partial \mathbf{q}_M} \end{bmatrix} \in \mathbb{R}^{(r_{\mathcal{D}} + n_S) \times r_{\mathcal{D}}},$$

where n_S denotes the number of slave amplitudes. The corresponding decoder Jacobian is therefore

$$\frac{\partial \mathcal{D}}{\partial \mathbf{q}_M} = \Phi \frac{\partial \boldsymbol{\tau}}{\partial \mathbf{q}_M}.$$

The second derivative of $\boldsymbol{\tau}$ is a third-order tensor:

$$\frac{\partial^2 \tau_i}{\partial q_{Mj} \partial q_{Mk}}, \quad i = 1, \dots, r_{\mathcal{D}} + n_S, \quad j, k = 1, \dots, r_{\mathcal{D}}.$$

In the implementation, this tensor is stored as `d2tau_dqM2(i,j,k)`. Since the master block $\mathbf{A}_M \mathbf{q}_M$ is linear, its second derivative is zero; only the slave block contributes, namely

$$\frac{\partial^2 \tau}{\partial q_M^2} = \begin{bmatrix} \mathbf{0} \\ \frac{\partial^2 \mathcal{N}_S}{\partial q_M^2} \end{bmatrix}.$$

6.4 Regression by Radial basis functions

The remaining ingredient in the decoder is the nonlinear closure $\mathbf{q}_S \approx \mathcal{N}_S(\mathbf{q}_M)$. Once the master coordinates have been fixed, this closure is constructed from the training pairs $(\mathbf{q}_{Mj}, \mathbf{q}_{Sj})$ obtained in Section 6.3. Several choices are possible. Deep feed-forward neural networks are now widely used in nonlinear manifold ROMs, particularly in problems involving high-dimensional latent spaces or strongly nonlinear manifold geometries [4]. In the present problem, however, the latent space is only two-dimensional and is approximately identified with the structured macroscopic parameter grid. This makes simpler differentiable interpolants attractive, including tensor-product spline constructions on Cartesian grids and radial basis functions. We adopt the latter option.

The first stage is to subsample the snapshots⁶. In this problem, since the latent coordinates are approximately aligned with the Cartesian macroscopic parameter grid, the subsampling can be carried out simply at the grid level by selecting one point every prescribed number of rows and columns. Let $\{\mathbf{q}_{Mk}^c\}_{k=1}^{n_{\mathcal{M}}}$ denote the resulting set of retained RBF centers, where $n_{\mathcal{M}} \leq n_s$ is the number of retained samples, and let $\boldsymbol{\ell} = (\ell_1, \ell_2)^T$ be the anisotropic length-scale vector. The Gaussian kernel is written as

$$\varphi(\mathbf{q}_M, \mathbf{q}_{Mk}^c) = \exp \left[- \left(\frac{q_{M,1} - q_{M,1,k}^c}{\ell_1} \right)^2 - \left(\frac{q_{M,2} - q_{M,2,k}^c}{\ell_2} \right)^2 \right]. \quad (50)$$

For a new master point $\mathbf{q}_M$, let

$$\boldsymbol{\psi}_{\text{RBF}}(\mathbf{q}_M) = \left[\varphi(\mathbf{q}_M, \mathbf{q}_{M1}^c) \quad \cdots \quad \varphi(\mathbf{q}_M, \mathbf{q}_{Mn_{\mathcal{M}}}^c) \right]^T \in \mathbb{R}^{n_{\mathcal{M}}}$$

denote the vector of Gaussian kernel evaluations with respect to the retained RBF centers. The slave amplitudes are then approximated as

$$\mathcal{N}_S(\mathbf{q}_M) = \mathbf{s}_S \left(\boldsymbol{\alpha}_{\text{RBF}}^T \boldsymbol{\psi}_{\text{RBF}}(\mathbf{q}_M) + \boldsymbol{\beta}_{\text{RBF}}^T \mathbf{p}(\mathbf{q}_M) \right). \quad (51)$$

⁶The finite element trajectories are sampled densely in pseudo-time in order to ensure robust Newton convergence. Consecutive snapshots are therefore often highly correlated in master space, so using all of them as RBF centers would increase the online cost and may deteriorate the conditioning of the kernel matrix without providing significant additional information.

Here

$$\mathbf{s}_S = \text{diag}(s_1, \dots, s_{n_S}), \quad s_i = \left(\frac{1}{n_{\mathcal{M}}} \sum_{k=1}^{n_{\mathcal{M}}} \left(q_{S,i}^{(k)} \right)^2 \right)^{1/2},$$

contains characteristic RMS amplitudes of the slave coordinates, used to normalize the multi-output regression problem and improve conditioning. The matrix $\boldsymbol{\alpha}_{\text{RBF}} \in \mathbb{R}^{n_{\mathcal{M}} \times n_S}$ collects the RBF coefficients, whereas $\boldsymbol{\beta}_{\text{RBF}} \in \mathbb{R}^{n_p \times n_S}$ contains the coefficients of the optional polynomial correction $\mathbf{p}(\mathbf{q}_M) \in \mathbb{R}^{n_p}$ (both $\boldsymbol{\alpha}_{\text{RBF}}$ and $\boldsymbol{\beta}_{\text{RBF}}$ are obtained in the offline stage, as explained in the following). In the present implementation, this correction may be absent, constant, or affine; in the affine case,

$$\mathbf{p}(\mathbf{q}_M) = [1 \quad q_{M,1} \quad q_{M,2}]^T.$$

6.4.1 Determination of RBFs parameters in the offline stage

Let

$$\mathbf{Q}_S = [\mathbf{q}_{S1} \quad \mathbf{q}_{S2} \quad \dots \quad \mathbf{q}_{Sn_{\mathcal{M}}}] \in \mathbb{R}^{n_S \times n_{\mathcal{M}}}$$

collect the slave amplitudes associated with the retained RBF centers. The corresponding normalized slave-coordinate matrix is

$$\hat{\mathbf{Q}}_S = \mathbf{s}_S^{-1} \mathbf{Q}_S \in \mathbb{R}^{n_S \times n_{\mathcal{M}}}.$$

The RBF coefficients are obtained offline by solving

$$\begin{bmatrix} \mathbf{K}_{\text{RBF}} + \lambda \mathbf{I} & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{\alpha}_{\text{RBF}} \\ \boldsymbol{\beta}_{\text{RBF}} \end{bmatrix} = \begin{bmatrix} \hat{\mathbf{Q}}_S^T \\ \mathbf{0} \end{bmatrix}, \quad (52)$$

where λ is a Tikhonov regularization parameter. The matrix $\mathbf{K}_{\text{RBF}} \in \mathbb{R}^{n_{\mathcal{M}} \times n_{\mathcal{M}}}$ is the Gaussian kernel matrix evaluated at the retained centers,

$$(\mathbf{K}_{\text{RBF}})_{ij} = \varphi(\mathbf{q}_{Mi}^c, \mathbf{q}_{Mj}^c), \quad i, j = 1, \dots, n_{\mathcal{M}},$$

and $\mathbf{P}_{\text{RBF}} \in \mathbb{R}^{n_{\mathcal{M}} \times n_p}$ is the polynomial matrix,

$$\mathbf{P}_{\text{RBF}} = \begin{bmatrix} \mathbf{p}(\mathbf{q}_{M1}^c)^T \\ \mathbf{p}(\mathbf{q}_{M2}^c)^T \\ \vdots \\ \mathbf{p}(\mathbf{q}_{Mn_{\mathcal{M}}}^c)^T \end{bmatrix}.$$

All slave outputs are fitted simultaneously using the same kernel matrix.

The differentiability of the RBF map is essential because the tangent matrix of the decoder requires $\partial \mathcal{N}_S / \partial \mathbf{q}_M$, and Newton-type procedures may also require second derivatives. Let

$$\mathbf{D}_{\Psi}(\mathbf{q}_M) := \frac{\partial \boldsymbol{\psi}_{\text{RBF}}}{\partial \mathbf{q}_M}(\mathbf{q}_M) \in \mathbb{R}^{n_{\mathcal{M}} \times r_{\mathcal{D}}}.$$

For the Gaussian kernel in Eq. 50, its entries are

$$(\mathbf{D}_{\Psi}(\mathbf{q}_M))_{ka} = \frac{\partial \varphi(\mathbf{q}_M, \mathbf{q}_{Mk}^c)}{\partial q_{M,a}} = -2 \frac{q_{M,a} - q_{M,a,k}^c}{\ell_a^2} \varphi(\mathbf{q}_M, \mathbf{q}_{Mk}^c), \quad a = 1, \dots, r_{\mathcal{D}}.$$

Therefore, differentiating Eq. 51 gives

$$\frac{\partial \mathcal{N}_S}{\partial \mathbf{q}_M}(\mathbf{q}_M) = \mathbf{s}_S \left[\boldsymbol{\alpha}_{\text{RBF}}^T \mathbf{D}_{\Psi}(\mathbf{q}_M) + \boldsymbol{\beta}_{\text{RBF}}^T \frac{\partial \mathbf{p}}{\partial \mathbf{q}_M}(\mathbf{q}_M) \right] \in \mathbb{R}^{n_S \times r_{\mathcal{D}}}. \quad (53)$$

The second derivative is a third-order tensor. For each center k , the Hessian of the Gaussian kernel is

$$\frac{\partial^2 \varphi(\mathbf{q}_M, \mathbf{q}_{Mk}^c)}{\partial q_{M,a} \partial q_{M,b}} = \left[4 \frac{(q_{M,a} - q_{M,a,k}^c)(q_{M,b} - q_{M,b,k}^c)}{\ell_a^2 \ell_b^2} - 2 \frac{\delta_{ab}}{\ell_a^2} \right] \varphi(\mathbf{q}_M, \mathbf{q}_{Mk}^c),$$

where δ_{ab} is the Kronecker delta. Hence, for each slave component $i = 1, \dots, n_S$,

$$\frac{\partial^2 (\mathcal{N}_S)_i}{\partial q_{M,a} \partial q_{M,b}} = s_i \left[\sum_{k=1}^{n_M} (\alpha_{\text{RBF}})_{ki} \frac{\partial^2 \varphi(\mathbf{q}_M, \mathbf{q}_{Mk}^c)}{\partial q_{M,a} \partial q_{M,b}} + \sum_{m=1}^{n_p} (\beta_{\text{RBF}})_{mi} \frac{\partial^2 (\mathbf{p})_m}{\partial q_{M,a} \partial q_{M,b}} \right]. \quad (54)$$

In the affine polynomial case used here, the second derivative of $\mathbf{p}$ is zero. Thus the Hessian tensor is entirely determined by the Gaussian kernels. In the implementation, the Jacobian is stored as an $n_S \times r_{\mathcal{D}}$ matrix and the second derivative as an $n_S \times r_{\mathcal{D}} \times r_{\mathcal{D}}$ array, with the last two indices corresponding to the two differentiation directions in master space.

The RBF closure $\mathbf{q}_S = \mathcal{N}_S(\mathbf{q}_M)$ is constructed offline in `RegressionViaRBF_master_slave.m`. The routine uses the master and slave training matrices, optionally performs structured subsampling of the two-dimensional master grid, computes anisotropic Gaussian length scales, normalizes each slave amplitude by its RMS value, and solves a regularized augmented RBF system. This is the implementation counterpart of Eqs. 50–52. Online evaluation is performed in `EvalSlaveRBF_Online.m`, which returns $\mathcal{N}_S(\mathbf{q}_M)$, its Jacobian $\partial \mathcal{N}_S / \partial \mathbf{q}_M$, and its second derivative tensor $\partial^2 \mathcal{N}_S / \partial \mathbf{q}_M^2$. This function is called by `EvalTauSlaveRBF_Online.m`, where the full ROM-coordinate map $\boldsymbol{\tau}(\mathbf{q}_M)$ and its derivatives are assembled. The structured subsampling of the master space is implemented in `SubsamplingMasterSpace.m`. Rather than selecting centers randomly, the routine exploits the Cartesian structure of the parameter grid and retains one point every prescribed number of rows and columns. This reduces the number of RBF centers, improves the conditioning of the Gaussian kernel matrix, and avoids excessive clustering of highly correlated snapshots originating from densely sampled Newton trajectories.

6.5 Numerical assessment

6.5.1 Numerical characterization of the manifold solution

We now characterize the nonlinear-manifold approximation constructed for the metamaterial cell. The starting point is the same displacement snapshot matrix used in the standard HROM, and the same reference tolerance $\varepsilon_d^{\text{SVD}} = 10^{-6}$. Thus, the initial linear basis is

$$\boldsymbol{\Phi} \in \mathbb{R}^{3018 \times 52}.$$

The purpose of the manifold construction is not to use these 52 coordinates as independent unknowns, but to identify a two-dimensional master parameterization and to reconstruct the remaining amplitudes through the nonlinear closure $\mathbf{q}_S \approx \mathcal{N}_S(\mathbf{q}_M)$.

The least-squares problem (38) gives a relative residual equal to 3.09×10^{-7} . This confirms that, for the present dataset, the master coordinates $\mathbf{q}_M$ induced by the master basis $\boldsymbol{\Phi}_M$ can be identified with the prescribed macroscopic input vector $\boldsymbol{\mu}$, according to Eq. (39) (i.e., $\mathbf{q}_M \approx \boldsymbol{\mu}$). Hence, this case falls within the input-driven setting anticipated in Remark 5: the latent dimension is fixed by the two independent macroscopic inputs, namely $r_{\mathcal{D}} = n_{\mu} = 2$.

The deformed shapes of the resulting master modes $\boldsymbol{\Phi}_{M1}$ and $\boldsymbol{\Phi}_{M2}$ are shown in Fig. 10. They should not be interpreted in the same way as the leading POD modes of Fig. 6. Although the first mode displays a deformation pattern broadly associated with horizontal compression/stretching, and the second one contains a shear-like signature, their main role is not modal energy ordering. Rather, they define the physical displacement directions whose amplitudes are placed in direct correspondence with the macroscopic input coordinates, up to the scaling matrix (obtained in turn from Eq. 41)

$$\mathbf{A}_M = 10^{-1} \begin{bmatrix} -3.9962 & 0.21588 \\ -0.34716 & -6.4264 \end{bmatrix}.$$

The remaining $r - r_{\mathcal{D}} = 52 - 2 = 50$ modes span the slave space, i.e. the orthogonal complement of the master subspace inside $\text{Col}(\Phi)$. The deformed shapes of some of such slave modes are shown in Fig. 11 for illustration purposes.

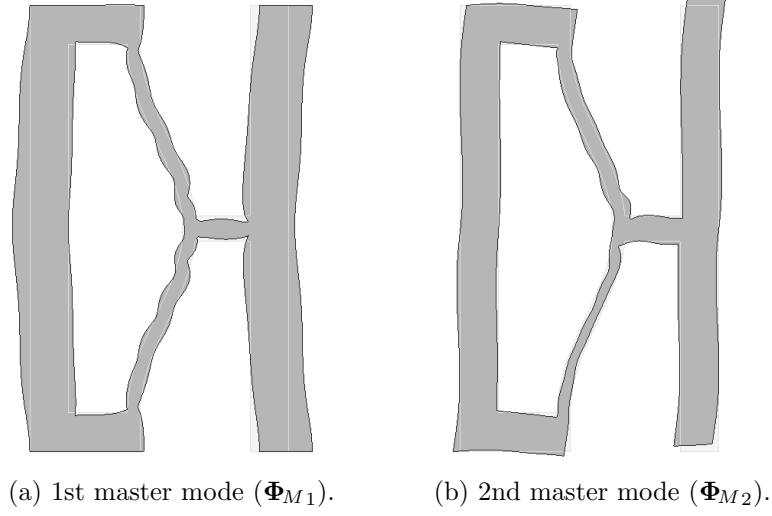


Figure 10: Master modes obtained from the least-squares identification of the two-dimensional latent space. The modes are not selected by POD energy ordering, but by their ability to induce amplitudes aligned with the macroscopic input coordinates.

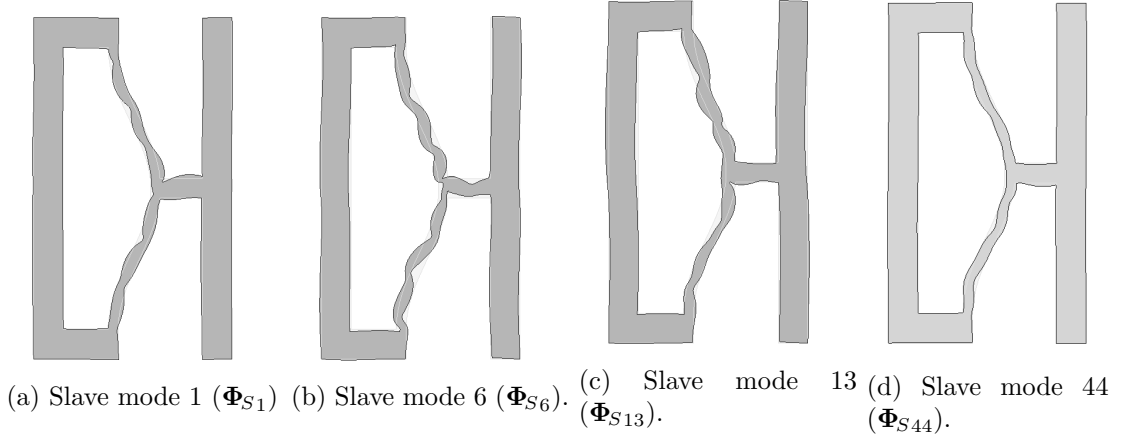


Figure 11: Slave modes Φ_{S1} , Φ_{S6} , Φ_{S13} and Φ_{S44} . The amplitudes of these modes are reconstructed through the nonlinear closure $\mathbf{q}_{Sj} \approx \mathcal{N}_{Sj}(\mathbf{q}_M)$.

The closure is built with the RBF approximation described in Section 6.4. The full dataset contains 4851 training pairs, from which $n_{\mathcal{M}} = 1540$ centers are retained after structured subsampling. The subsampling frequency is $[3, 1]$, so that every third point is retained along the first master direction, while all points are retained along the second one. The length-scale factors are chosen⁷ as

$$\ell = \begin{bmatrix} 3.3013 \times 10^{-2} \\ 1.5789 \times 10^{-2} \end{bmatrix}.$$

Tikhonov parameter is set to $\lambda = 10^{-10}$, and the polynomial correction is parabolic.

⁷These values are obtained from the user-prescribed factors $[3, 3]$ by multiplying the average spacing between neighboring retained RBF centers along each master direction. Thus, the parameters 3 and 3 do not define the length scales directly, but rather the number of characteristic point spacings used to determine the effective Gaussian support radius in each direction.

With these parameters, the relative prediction error on the retained RBF centers is 4.30×10^{-6} , while the reconstruction error over the full dataset is 1.03×10^{-4} . This distinction is relevant: the first number measures the interpolation quality on the subsampled centers, whereas the second one measures the ability of the RBF closure to reconstruct all slave amplitudes over the original dataset.

For comparison, reducing the length-scale factors by a factor $2/3$ gives a much smaller residual on the retained centers, with prediction error 1.37×10^{-10} , but increases the full-dataset reconstruction error to 4.86×10^{-4} . Thus, although the shorter length scale interpolates the retained centers more tightly, it produces a less accurate reconstruction away from them. The first choice is therefore retained, since it provides the better global reconstruction over the complete training set and a smoother closure in master space. Figures 12 and 13 give a concrete visualization of the closure relation $\mathbf{q}_S \approx \mathcal{N}_S(\mathbf{q}_M)$. The black markers indicate the retained RBF centers. The plotted surfaces show that the dependence on the two master coordinates is smooth and single-valued over the sampled domain. This supports the chosen parameterization: once the master coordinates are identified with the macroscopic inputs, the remaining slave amplitudes can be regarded as regular functions of $\mathbf{q}_M$.

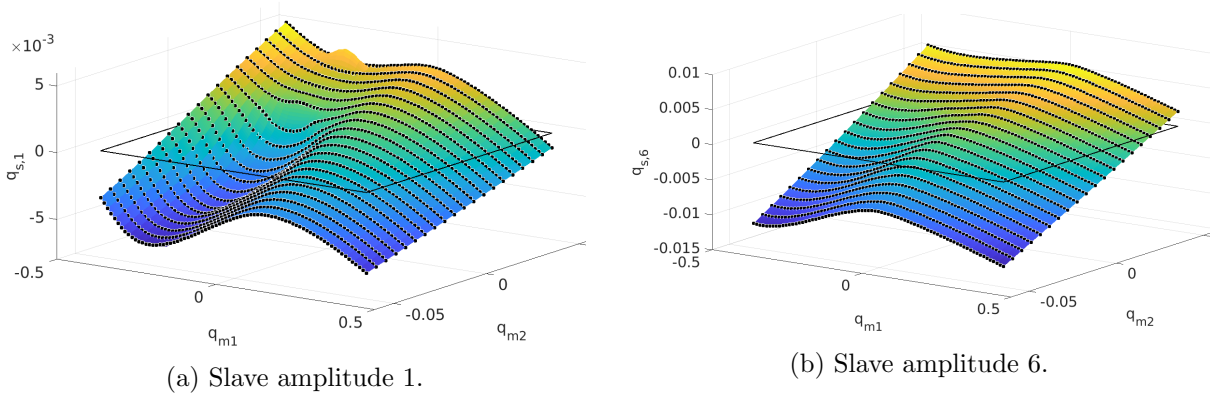


Figure 12: RBF reconstruction of slave amplitudes $\mathbf{q}_{S_1}$ and $\mathbf{q}_{S_6}$ over the two-dimensional master space. The black markers denote the retained RBF centers.

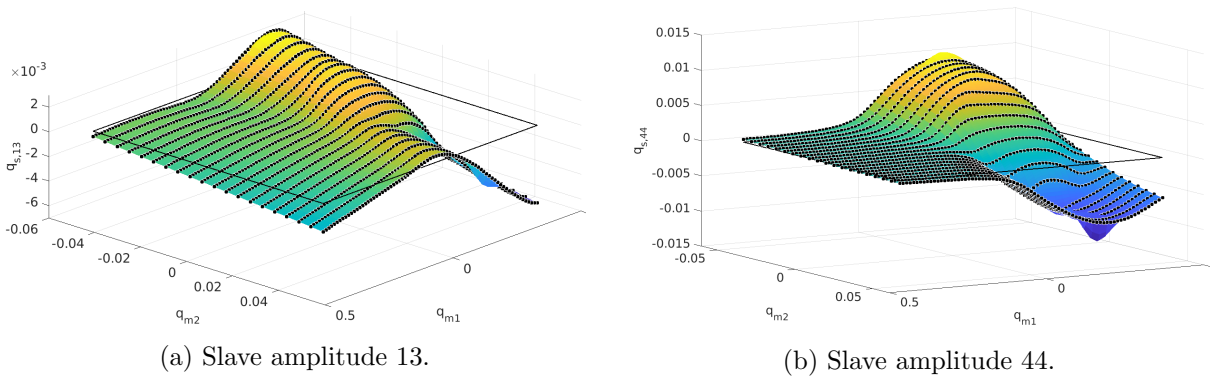


Figure 13: Additional examples of RBF-reconstructed slave amplitudes.

The fixed-weight manifold HROM for the metamaterial cell is launched from `Run_HOMOG_2p.m`. The input file is `Data_M3_hyperSTD.m`. The manifold option is activated with `DATA.interp.METHOD.SELECT-REFERENCE.MODE = 'TWO-VARIABLES'`. The fixed-weight hyperreduction setting is selected with `DATAoffline.Hyperreduction.METHOD-projection = 'STANDARD'`. The identification $\mathbf{q}_M \equiv \boldsymbol{\mu}$ is imposed through `DATA.interp.MethodSelectMasterModes_2var = 'LeastSquaresParametricDomain'`. Since only the output cubature is used in this case, the flags are set as `DATAoffline.ECMforReactionForces = true` and `DATAoffline.ECMforEquilibrium = false`. The offline operations are carried out in `OFFLINE_STAGE_nonECM_2var.m`. The least-squares construction of $\mathbf{T}_M$, $\boldsymbol{\Phi}_c$, $\boldsymbol{\Phi}_M$, and $\mathbf{A}_M$ is implemented in `MasterModes2var_LSqares.m`. The construction of $\boldsymbol{\Phi}_S$, the computation of $\mathbf{q}_M$

and $\mathbf{q}_S$, and the regression of $\mathbf{q}_S = \mathcal{N}_S(\mathbf{q}_M)$ are performed in `Determine_qinf.qsup.2var.m`.

The initial SVD basis used for the manifold construction is computed in `Determine_qinf.qsup.2var.m`. The least-squares identification of the master coordinates is performed in `MasterModes2var.LSquares.m`. For the present run, the relative least-squares residual `ErrorLS.ad` is 3.0888×10^{-7} . The matrix `Am` returned by the routine corresponds to the scaling matrix $\mathbf{A}_M$ in Eq. 41.

The RBF closure is constructed in `RegressionViaRBF_master_slave.m`. The parameters used here are `SubSamplingMasterSpaceFrequency` = [3,1], `c_factor_length.parameter` = [3,3], `PolynomialCorrectionNumberOfMonomials` = 2, and `lambdaREGULARIZATION` = 1e-10. The length scales are computed in `LengthParametersRBF.m`, giving $\ell = (0.033013, 0.015789)^T$. With these parameters, the prediction error on the retained centers is 4.2979×10^{-6} and the full-dataset reconstruction error is 1.0252×10^{-4} .

6.5.2 Fixed-weight ECM on the nonlinear manifold

With the manifold coordinates and the slave closure in place, the next step is the construction of the fixed cubature rule. The procedure follows the same ECM logic used in Section 5.5, but the objects to be integrated are now those induced by the nonlinear manifold. Since $\mathbf{q}_M \approx \boldsymbol{\mu}$ in the present input-driven setting, two possibilities arise. One may retain the projected manifold equilibrium equation (29), which leads to a system of $r_{\mathcal{D}} = 2$ nonlinear equations, and use the same cubature rule also for the homogenized stress in Eq. (30). Alternatively, one may exploit the fact that the master coordinates are already prescribed by the macroscopic input and construct a rule only for the output functional.

We first consider the former case, in which both the manifold-projected equilibrium block and the stress-output block are included. For each retained training state, we assemble the concatenated matrix

$$\mathbf{a}_{\mathcal{DP}}(\mathbf{q}_{Mj}; \boldsymbol{\mu}_j) = [\mathbf{a}_{\mathcal{D}}(\mathbf{q}_{Mj}; \boldsymbol{\mu}_j) \quad \mathbf{a}_P(\mathbf{q}_{Mj}; \boldsymbol{\mu}_j)] \in \mathbb{R}^{n_{\text{gp}} \times (r_{\mathcal{D}} + n_{\sigma})}. \quad (55)$$

In the present problem, $r_{\mathcal{D}} + n_{\sigma} = 2 + 4 = 6$. The matrix given to the SVD is therefore obtained by collecting these blocks over the $n_{\mathcal{M}} = 1540$ retained training states,

$$[\mathbf{a}_{\mathcal{DP}}(\mathbf{q}_{M1}; \boldsymbol{\mu}_1) \quad \cdots \quad \mathbf{a}_{\mathcal{DP}}(\mathbf{q}_{Mn_{\mathcal{M}}}; \boldsymbol{\mu}_{n_{\mathcal{M}}})] \in \mathbb{R}^{5260 \times 9240}.$$

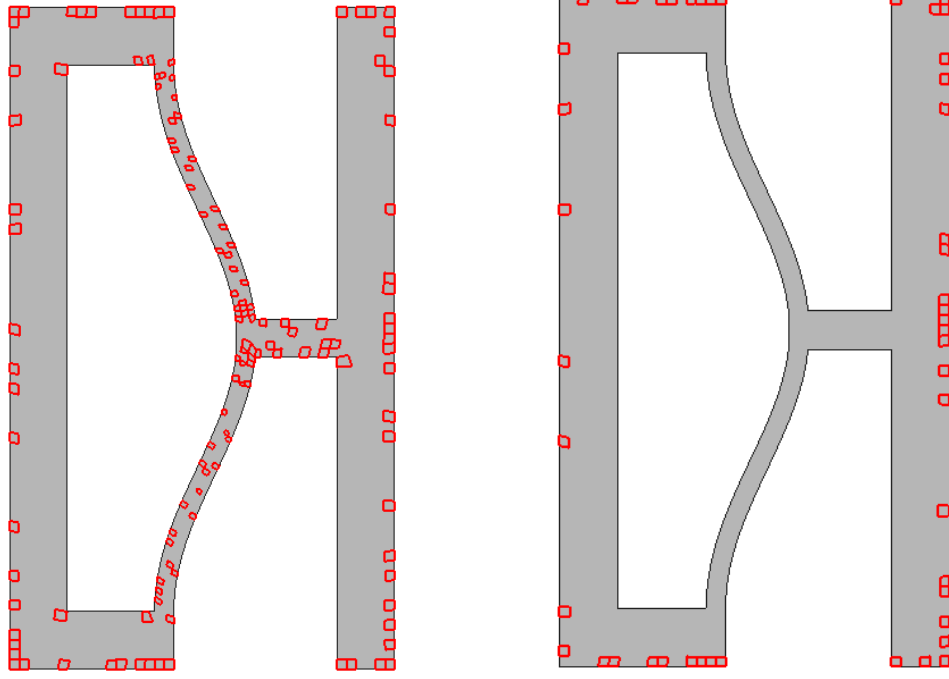
The states used in this construction are consistent with the nonlinear manifold: each FE snapshot is first encoded through Eq. (49), then decoded through Eq. (36), and finally lifted to the full nodal space by Eq. (1). The Gauss-point strains, stresses, internal-force densities and output quantities are evaluated from this decoder-reconstructed displacement field.

Using an SVD tolerance of 10^{-4} , the concatenated equilibrium–output integrand matrix yields 145 left singular vectors. After the additional constant-integration constraint is imposed, the ECM basis contains 146 vectors, and the cubature rule therefore selects 146 Gauss points. These points are contained in 136 elements.

The second case exploits the input-driven character of the present example more directly. If $\mathbf{q}_M$ is identified with the prescribed macroscopic input, the online equilibrium solve in $\mathbf{q}_M$ is no longer needed for the purpose of evaluating the homogenized response. The cubature rule may then be built only from the stress-output block. Each retained state contributes only $n_{\sigma} = 4$ columns, so that the SVD is applied to a matrix of size

$$5260 \times (n_{\sigma} n_{\mathcal{M}}) = 5260 \times 6160.$$

For the same SVD tolerance, this output-only construction gives 54 cubature points, distributed over 48 elements.



(a) Equilibrium and PK1 output: 146 points in 136 elements.

(b) PK1 output only: 54 points in 48 elements.

Figure 14: Selected Gauss points for the fixed-weight manifold ECM. Compared with the standard HROM rule of Fig. 7, which required 1340 points, the reduction is substantial. The decrease is mainly due to the fact that the tangent space of the nonlinear manifold has only $r_{\mathcal{D}} = 2$ directions, instead of the $r = 52$ fixed linear directions of the standard ROM. When only the stress-output block is retained, the cubature rule must reproduce only the PK1 stress integrands, leading to an even smaller set of points.

The comparison with Fig. 7 is instructive. In the standard HROM, each training state contributes $r + n_{\sigma} = 56$ scalar integrand fields. In the manifold HROM, the equilibrium part contributes only $r_{\mathcal{D}} = 2$ tangent directions, so each state contributes $2 + 4 = 6$ fields when both equilibrium and output are included. The stresses themselves are not made simpler by the manifold construction; rather, the virtual strains used to test the internal force live in a two-dimensional tangent space that moves along the curved manifold. This strongly reduces the dimension of the internal-force density space seen by ECM. If the equilibrium block is removed altogether, only the four stress-output components remain, explaining the further decrease from 146 to 54 points.

The equilibrium-output case is run from `Run.HOMOG_2p.m` with input file `Data.M4_hyperSTD.m`. The flags are `DATAoffline.ECMforReactionForces = true` and `DATAoffline.ECMforEquilibrium = true`. The Newton tolerances for the manifold HROM solve are set to `TOL_FORCES_REL = 1e-10`, `TOL_DISPLAC_ABS = 1e-10`, and `NMAXiter = 100`.

Although the main text writes the homogenized PK1 stress through the volume average, the implementation uses the equivalent reaction-based route. The reaction-output operator is assembled in `DetermineModeReactionsOUTPUT_glo.m`, which calls `DetermineModeHomogViaREACT.m`. When the equilibrium and reaction-output blocks are concatenated, the reaction-output block is normalized by its Frobenius norm and rescaled with the Frobenius norm of the equilibrium block before the SVD, in order to balance the two contributions.

The integrand blocks are assembled in `OFFLINE_STAGE_nonECM_2var.m`. Only the subsampled snapshots retained as RBF centers are used; this is encoded through `DATA_interp.indices_subsampling_SNAPSHOTS`. For each retained state, the code stores the equilibrium and output contributions together in a single block of size $n_{gp} \times (r_{\mathcal{D}} + n_{\sigma})$, rather than storing $\mathbf{A}_{\mathcal{D}}$ and $\mathbf{A}_{\mathcal{P}}$ as two separate global matrices. This organization is immaterial for a fixed ECM rule, but it prepares the data layout used later by the manifold-adaptive-weight ECM.

6.5.3 Online assessment of the fixed-weight manifold HROM

As already noted, the fixed-weight manifold construction leads to two online strategies. In the first one, the master coordinates are identified directly with the prescribed macroscopic input, $\mathbf{q}_M \approx \boldsymbol{\mu}$, and the homogenized stress is obtained by direct evaluation of the decoder and of the output cubature rule., and the homogenized stress is obtained by direct evaluation of the decoder and of the output cubature rule. No reduced equilibrium problem is solved. For a new input $\boldsymbol{\mu}$, one sets $\mathbf{q}_M = \boldsymbol{\mu}$, evaluates⁸ $\mathbf{d}_L = \mathcal{D}(\mathbf{q}_M)$, and uses the lifting (1) only on the elements containing the selected cubature points. More precisely, if $\mathcal{E}_{\text{sel}}$ denotes the set of elements containing the selected Gauss points, only the restrictions

$$\mathbf{d}|_{\mathcal{E}_{\text{sel}}} = (\mathbf{A}\mathcal{D}(\mathbf{q}_M) + \bar{\mathbf{U}}\boldsymbol{\mu})|_{\mathcal{E}_{\text{sel}}}$$

are needed for the evaluation of the stresses entering the reduced homogenized-stress formula 30. Thus, the online evaluation is a direct functional evaluation on the selected elements, without pseudo-time continuation and without Newton iterations.

The second strategy keeps the projected manifold equilibrium equation (29). In this case, for each prescribed $\boldsymbol{\mu}$, the two master coordinates are obtained by solving a nonlinear system of size $r_{\mathcal{D}} = 2$. The residual is

$$\mathbf{R}_{\mathcal{D}}(\mathbf{q}_M; \boldsymbol{\mu}) = \sum_{g \in \mathcal{Z}_{\mathcal{D}}} \boldsymbol{\Phi}_{\mathcal{D}}(\mathbf{q}_M)^T \mathbf{r}_g^{\text{FE}}(\mathbf{q}_M; \boldsymbol{\mu}) w_g^{\mathcal{D}}.$$

Its Newton linearization leads to the tangent matrix (that is, the Jacobian of the reduced nonlinear system with respect to the master coordinates), which contains two distinct contributions. Applying the chain rule gives

$$\mathbf{K}_{\mathcal{D}}(\mathbf{q}_M; \boldsymbol{\mu}) = \mathbf{K}_{\mathcal{D}}^{\text{std}}(\mathbf{q}_M; \boldsymbol{\mu}) + \mathbf{K}_{\mathcal{D}}^{\text{curv}}(\mathbf{q}_M; \boldsymbol{\mu}), \quad (56)$$

where the standard tangent contribution is

$$\mathbf{K}_{\mathcal{D}}^{\text{std}} = \sum_{g \in \mathcal{Z}_{\mathcal{D}}} \boldsymbol{\Phi}_{\mathcal{D}}(\mathbf{q}_M)^T \mathbf{K}_g^{\text{FE}}(\mathbf{q}_M; \boldsymbol{\mu}) \boldsymbol{\Phi}_{\mathcal{D}}(\mathbf{q}_M) w_g^{\mathcal{D}}, \quad (57)$$

with $\mathbf{K}_g^{\text{FE}}$ denoting the constrained Gauss-point tangent associated with $\mathbf{r}_g^{\text{FE}}$. The additional term is caused by the dependence of the tangent basis $\boldsymbol{\Phi}_{\mathcal{D}}(\mathbf{q}_M)$ on the state. Componentwise,

$$(\mathbf{K}_{\mathcal{D}}^{\text{curv}})_{ab} = \sum_{g \in \mathcal{Z}_{\mathcal{D}}} \left[\frac{\partial^2 \mathcal{D}}{\partial q_{M,a} \partial q_{M,b}}(\mathbf{q}_M) \right]^T \mathbf{r}_g^{\text{FE}}(\mathbf{q}_M; \boldsymbol{\mu}) w_g^{\mathcal{D}}, \quad a, b = 1, \dots, r_{\mathcal{D}}. \quad (58)$$

This residual-weighted curvature term is absent in a linear ROM because $\boldsymbol{\Phi}_{\mathcal{D}}$ is then constant. It is also commonly neglected in Gauss–Newton-type linearizations of nonlinear projection-based ROMs when the residual is assumed small near convergence [3, 2]. In the present implementation it is retained whenever the second derivative of the decoder is available.

The comparison with the standard HROM is reported in Table 2. The errors are averaged over the two training trajectories, as in Table 1. In the manifold cases, the displacement error is not a pure consistency check: the RBF closure is trained only on the $n_{\mathcal{M}} = 1540$ retained centers, whereas the reported errors are evaluated over the full set of snapshots.

⁸From an implementation viewpoint, the same computational pipeline as in the nonlinear manifold solve is retained. The only difference is that the tentative value of the master coordinates is prescribed directly from the macroscopic input, rather than initialized from the previous pseudo-time step and subsequently corrected through Newton iterations. The iterative stage is then effectively bypassed by imposing convergence tolerances that are satisfied immediately.

Table 2: Online accuracy of the fixed-weight manifold HROM compared with the standard HROM. The manifold models use $r_{\mathcal{D}} = 2$ master coordinates and $r - r_{\mathcal{D}} = 50$ slave coordinates.

Model	Master coords.	Slave coords.	n_{ECM}	e_d	e_P
ROM, full Gauss rule	52	0	5260	2.52×10^{-6}	5.05×10^{-5}
HROM, ECM	52	0	1340	1.85×10^{-5}	7.13×10^{-5}
M-HROM, equil. + PK1	2	50	146	9.64×10^{-5}	8.81×10^{-3}
M-HROM, PK1	2	50	54	7.87×10^{-5}	2.98×10^{-4}

The output-only strategy gives the best stress accuracy among the two manifold variants. This is a direct consequence of the exceptionally small residual in the least-squares identification $\mathbf{q}_M = \mathbf{T}_M \mathbf{q} \approx \boldsymbol{\mu}$, which in the present case is of the order of 10^{-7} (as we saw in Section 6.5.1). In other words, the macroscopic input variables already provide an extremely accurate parametrization of the solution manifold. Under these circumstances, solving an additional projected equilibrium problem in the latent space brings essentially no additional information and may even introduce small deviations from the input-driven manifold representation. Accordingly, the most accurate strategy is simply to evaluate directly the decoder and the reduced stress functional from the prescribed input coordinates, without solving any nonlinear system.

It should be emphasized that this situation is highly problem-dependent. The present example corresponds to a static hyperelastic homogenization problem without internal history variables, for which the displacement manifold is almost entirely parametrized by the macroscopic loading variables. As will be shown later for history-dependent constitutive behavior, such as damage models, this direct identification $\mathbf{q}_M \approx \boldsymbol{\mu}$ no longer holds, and solving the projected equilibrium equations becomes unavoidable.

Naturally, the manifold approximation introduces an additional source of error with respect to the standard HROM. This is reflected in the increase of the relative homogenized-stress error from approximately $7 \cdot 10^{-5}$ in the standard HROM to roughly $3 \cdot 10^{-4}$ in the output-only manifold approximation. Nevertheless, this deterioration remains moderate: the loss is approximately one order of magnitude in accuracy, while the reduction in online complexity is dramatically larger. Indeed, the number of required Gauss-point evaluations decreases from the original $n_{\text{gp}} = 5260$ points to only 54, that is, almost a two-order-of-magnitude reduction. Moreover, this reduction is achieved without solving any nonlinear equilibrium problem during the online stage: the response is obtained directly by evaluating the decoder and the reduced stress functional.

At the same time, the present results also reveal the remaining limitation of the fixed-weight paradigm. Despite the dramatic reduction from 5260 to 54 integration points, the resulting cubature rule still requires a number of points approximately 25 times larger than the number of latent coordinates ($r_{\mathcal{D}} = 2$). This indicates that the complexity of the cubature rule does not scale directly with the dimension of the latent manifold itself. This disparity can be expected to become more pronounced as the dimension of the input parameter space increases. The adaptive ECM introduced next is motivated precisely by this observation: rather than enforcing a single cubature rule over the entire manifold, the integration weights are allowed to evolve with the position on the manifold.

The Newton strategy is implemented in `NewtonRapshonStaticLarge_manifoldMD.m`. The residual and tangent are assembled from `ResidualFromDisplacementsVAR.m`. The additional curvature contribution corresponds to the code block in which `tauNONder2.q` is contracted with `VAR.RESID_extend`; this is the implementation counterpart of Eq. 58. The equilibrium–output run uses `Data_M4_hyperSTD.m`, whereas the output-only evaluation uses `Data_M3_hyperSTD.m`.

7 Manifold-adaptive weights ECM

7.1 Motivation

The fixed-weight manifold HROM developed in the previous section already achieves a substantial reduction of the integration cost. In particular, the output-only formulation reduces the cubature rule to only 54 Gauss points, while preserving an accurate approximation of the homogenized stresses. Nevertheless, this result also reveals an important limitation of the fixed cubature paradigm. Although the nonlinear manifold is parametrized here by only $r_{\mathcal{D}} = 2$ latent coordinates, the number of integration points remains significantly larger than the intrinsic dimension of the manifold.

This observation is not accidental. The nonlinear manifold ROM recognizes that the solution snapshots evolve on a curved low-dimensional manifold embedded in the high-dimensional displacement space (i.e., $\mathbf{d}_L \approx \mathcal{D}(\mathbf{q}_M)$). However, the cubature rule introduced in Eqs. (29)–(30) does not exploit this geometric structure locally. The same set of points $\mathcal{Z}_{\mathcal{D}}$ and the same weights $w_g^{\mathcal{D}}$ are used for all positions on the manifold, independently of the current latent coordinate $\mathbf{q}_M$.

The origin of this limitation becomes clearer when examining the construction of the ECM basis. For each sampled manifold state $(\mathbf{q}_{Mj}, \boldsymbol{\mu}_j)$, the corresponding equilibrium and stress-output contributions are assembled through the block matrix introduced in Eq. (55). Each sampled state therefore contributes $n_{\mathcal{DP}} := r_{\mathcal{D}} + n_{\sigma}$ scalar Gauss-point fields. In the present example, $n_{\mathcal{DP}} = 2 + 4 = 6$. The global matrix used to construct the fixed cubature rule is then obtained by concatenating all these local blocks over the sampled manifold, as expressed in Eq. (33).

This concatenation is the algebraic manifestation of a global integration strategy. The cubature weights are required to satisfy, approximately, one single set of linear equality constraints associated with the union of all sampled integrand configurations over the manifold. In other words, although the integrands evolve continuously with $\mathbf{q}_M$, the cubature operator itself remains fixed.

The modification proposed in this section can be stated directly at the level of the reduced integration rule. Instead of using the fixed weights $w_g^{\mathcal{D}}$ appearing in Eqs. (29)–(30), we seek a reduced rule of the form

$$\sum_{g \in \mathcal{Z}_{\omega}} \mathbf{r}_g^{\mathcal{D}}(\mathbf{q}_M; \boldsymbol{\mu}) \omega_g(\mathbf{q}_M) = \mathbf{0}, \quad (59)$$

for the manifold-projected equilibrium equations, and

$$\mathbf{P}^{\text{macro}} \approx \frac{1}{|\Omega_0|} \sum_{g \in \mathcal{Z}_{\omega}} \mathbf{P}_g(\mathbf{q}_M; \boldsymbol{\mu}) \omega_g(\mathbf{q}_M) \quad (60)$$

for the homogenized stress output. The set $\mathcal{Z}_{\omega}$ is common to all sampled manifold states and contains

$$n_{\omega} := |\mathcal{Z}_{\omega}|$$

integration points. What changes from one point of the manifold to another is not the support, but the value of the weights.

Thus, each retained Gauss point $g \in \mathcal{Z}_{\omega}$ is assigned a scalar field over the latent space,

$$\omega_g : \mathbb{R}^{r_{\mathcal{D}}} \rightarrow \mathbb{R}_+, \quad \mathbf{q}_M \mapsto \omega_g(\mathbf{q}_M).$$

Equivalently, the cubature rule is described by the manifold-dependent vector

$$\boldsymbol{\omega}(\mathbf{q}_M) = [\omega_{g_1}(\mathbf{q}_M), \dots, \omega_{g_{n_{\omega}}}(\mathbf{q}_M)]^T, \quad \mathcal{Z}_{\omega} = \{g_1, \dots, g_{n_{\omega}}\}.$$

This is the sense in which the method is called a manifold-adaptive weights ECM, or MAW-ECM: the integration points remain fixed after the offline stage, whereas the associated ECM weights adapt to the position on the nonlinear manifold.

The above problem conceals several challenging tasks from the optimization point of view, namely:

1. The final number of integration points (n_ω) should be as small as possible.
2. The reduced cubature rule must reproduce, up to a prescribed tolerance, the linear equality constraints associated with the manifold-projected equilibrium equations 59 and/or the output of interest (homogenized stress in this case, Eq. 60). In practice, each sampled manifold state contributes $n_{\mathcal{D}P} = r_{\mathcal{D}} + n_\sigma$ scalar integration conditions, and the adaptive weights must satisfy all these local exactness conditions with an user-prescribed accuracy throughout the sampled latent domain.
3. The n_ω weights must remain nonnegative throughout the sampled latent domain, while preserving the exact total volume, i.e., $\sum_{g \in \mathcal{Z}_\omega} \omega_g(\mathbf{q}_M) = |\Omega_0|$.
4. Finally, each scalar field $\omega_g(\mathbf{q}_M)$ should vary as smoothly as possible over the manifold. Indeed, the adaptive weights will ultimately be represented through a regression procedure over the latent domain (for instance using RBF interpolation or neural networks), starting from weight values computed at discrete sampled manifold states. Strong local oscillations or abrupt changes between neighboring states would therefore deteriorate both the regression accuracy of the weight fields and the robustness of the online stage when iterative solvers are required to solve the reduced equilibrium equations.

The rationale behind adaptive cubature weights is perhaps best understood on a simple analytical example, where the mechanism can be examined explicitly. Consider the family of monomials

$$f_q(x) = x^q, \quad x \in [0, 1], \quad q = 0, 1, \dots, m.$$

The exact integrals are

$$\int_0^1 f_q(x) dx = \frac{1}{q+1}.$$

If one seeks a fixed integration rule capable of reproducing exactly all these integrals simultaneously,

$$\int_0^1 f_q(x) dx = \sum_{g=1}^{n_{\text{int}}} f_q(x_g) \omega_g, \quad q = 0, 1, \dots, m,$$

then the number of integration points must increase with the number of monomials (as it happens in Gauss quadrature). However, if the weights are allowed to depend on the parameter q , the whole family can be integrated exactly using a single integration point located at $x = 1$, i.e., $n_\omega = 1$,

$$\int_0^1 f_q(x) dx = \omega(q) f_q(1), \quad \omega(q) = \frac{1}{q+1}.$$

Thus, instead of increasing the number of integration points in order to capture the whole family with one fixed rule, the complexity is transferred to a smooth parameter-dependent weight field.

7.2 Initialization and local integration constraints

The requirements 1–4 will be formulated in the sequel as a constrained optimization problem. Before doing so, however, it is convenient to discuss the equality constraints entering the formulation and the accuracy with which they are required to be satisfied. Firstly, we anticipate that the proposed construction follows an iterative pruning strategy initialized from the fixed-weight

ECM rule obtained from Problem (34), whose sparse solution is $\mathbf{w}^{\mathcal{D}} \in \mathbb{R}^{n_{\text{gp}}}$ and its support $\mathcal{Z}_{\mathcal{D}} := \text{supp}(\mathbf{w}^{\mathcal{D}}) \subset \{1, \dots, n_{\text{gp}}\}$. Accordingly, the support associated with the adaptive cubature rule will be sought within $\mathcal{Z}_{\mathcal{D}}$, i.e.,

$$\mathcal{Z}_{\omega} \subseteq \mathcal{Z}_{\mathcal{D}}.$$

Let $\mathbf{w}_{\mathcal{Z}_{\mathcal{D}}}^{\mathcal{D}} \in \mathbb{R}^{|\mathcal{Z}_{\mathcal{D}}|}$ denote the restriction of the sparse ECM weights $\mathbf{w}^{\mathcal{D}}$ to $\mathcal{Z}_{\mathcal{D}}$. For subsequent developments, and in order to keep the optimization formulation as general as possible, we introduce the notation

$$\boldsymbol{\omega}_0 := \mathbf{w}_{\mathcal{Z}_{\mathcal{D}}}^{\mathcal{D}}, \quad m_0 := |\mathcal{Z}_{\mathcal{D}}|.$$

where $|\mathcal{Z}_{\mathcal{D}}|$ denotes the cardinality of $\mathcal{Z}_{\mathcal{D}}$. Thus, $\boldsymbol{\omega}_0 \in \mathbb{R}^{m_0}$ represents the initial vector of candidate weights, while m_0 denotes the initial number of candidate integration points prior to the adaptive pruning process.

The fact that the fixed ECM rule provides the reference integration rule for MAW-ECM algorithm has important implications on the way the integration conditions associated with Eqs. (59)–(60) are interpreted. As discussed in Section 6.2, the state-wise equilibrium and output blocks $\mathbf{a}_{\mathcal{DP}}(\mathbf{q}_{M_j})$, $j = 1, \dots, n_{\mathcal{M}}$, are first collected over the sampled manifold and compressed through a truncated SVD. The left singular vectors retained in $\mathbf{U}$ define the integrand subspace on which the fixed ECM rule is exact. Thus, rather than requiring the reduced rule to reproduce the FE-quadrature integrals of all columns of the original local block $\mathbf{a}_{\mathcal{DP}}(\mathbf{q}_{M_j}) \in \mathbb{R}^{n_{\text{gp}} \times (r_{\mathcal{D}} + n_{\sigma})}$, we require exact reproduction only for their projection onto the retained local integrand subspace. To this end, we define

$$\mathbf{u}_{\mathcal{DP}}(\mathbf{q}_{M_j}) := \mathbf{U}\mathbf{U}^T \mathbf{a}_{\mathcal{DP}}(\mathbf{q}_{M_j}) \in \mathbb{R}^{n_{\text{gp}} \times r_c}, \quad j = 1, \dots, n_{\mathcal{M}}. \quad (61)$$

The fixed ECM rule then satisfies

$$\mathbf{u}_{\mathcal{DP}}(\mathbf{q}_{M_j})^T \mathbf{w}^{\mathcal{D}} = \mathbf{u}_{\mathcal{DP}}(\mathbf{q}_{M_j})^T \mathbf{w}^{\text{FE}}, \quad j = 1, \dots, n_{\mathcal{M}}. \quad (62)$$

These equalities define the local integration conditions inherited by the adaptive construction.

The projected blocks $\mathbf{u}_{\mathcal{DP}}(\mathbf{q}_{M_j})$ are not the only constraints entering the adaptive construction. Requirement 3 in Section 7.1 already imposes exact reproduction of the total volume, which may be regarded as a manifold-independent invariant of the cubature rule. More generally, one may wish to enforce additional invariant integration properties uniformly over the sampled manifold, such as exact reproduction of low-order geometric moments (e.g. inertia moments) or of reference integrands associated with the linear regime. Such constraints are not attached to any particular manifold state, but are instead required to hold for every adaptive cubature rule. We collect these state-independent exactness constraints in a block

$$\mathbf{U}_{\text{inv}} \in \mathbb{R}^{n_{\text{gp}} \times r_{\text{inv}}}.$$

(at minimum, $\mathbf{U}_{\text{inv}}$ contains the constant integrand $\mathbf{1} \in \mathbb{R}^{n_{\text{gp}}}$, i.e., an all-ones vector)

For each sampled point $\mathbf{q}_{M_j}$, the full-grid local constraint basis is obtained by combining the invariant block with the projected local integrand content,

$$\bar{\mathbf{U}}_j := \text{orth}([\mathbf{U}_{\text{inv}}, \mathbf{u}_{\mathcal{DP}}(\mathbf{q}_{M_j})]) \in \mathbb{R}^{n_{\text{gp}} \times r_c}, \quad j = 1, \dots, n_{\mathcal{M}}.$$

The orthogonalization removes redundant conditions between the invariant block and the local projected fields. The adaptive unknowns are initialized on the fixed ECM support $\mathcal{Z}_{\mathcal{D}}$. Therefore, the constraint basis used in the adaptive problem is the restriction of $\bar{\mathbf{U}}_j$ to the rows indexed by $\mathcal{Z}_{\mathcal{D}}$:

$$\mathbf{U}_j := \bar{\mathbf{U}}_j|_{\mathcal{Z}_{\mathcal{D}}} \in \mathbb{R}^{m_0 \times r_c}.$$

The corresponding local right-hand side inherited from the reference fixed rule is

$$\mathbf{b}_j := \mathbf{U}_j^T \boldsymbol{\omega}_0 \in \mathbb{R}^{r_c}.$$

Thus, at the sampled manifold state $\mathbf{q}_{Mj}$, the adaptive weights must satisfy

$$\mathbf{U}_j^T \boldsymbol{\omega}_j = \mathbf{b}_j, \quad j = 1, \dots, n_{\mathcal{M}}. \quad (63)$$

Here $\boldsymbol{\omega}_j \in \mathbb{R}^{m_0}$ denotes the adaptive weight vector associated with $\mathbf{q}_{Mj}$, still expressed on the initial candidate support $\mathcal{Z}_{\mathcal{D}}$. The subsequent pruning process will progressively eliminate entries of these vectors while preserving the same local equality constraints.

7.3 Sparsification problem

The local exactness conditions in (63) define, at each sampled manifold state $\mathbf{q}_M^{(j)}$, a feasible set of adaptive cubature weights. The unknowns of the problem are therefore not a single cubature vector, but the collection

$$\{\boldsymbol{\omega}_1, \boldsymbol{\omega}_2, \dots, \boldsymbol{\omega}_{n_{\mathcal{M}}}\},$$

where each

$$\boldsymbol{\omega}_j \in \mathbb{R}^{m_0}$$

contains the cubature weights associated with the sampled latent state $\mathbf{q}_M^{(j)}$, expressed on the initial candidate support $\mathcal{Z}_{\mathcal{D}}$.

Equivalently, the adaptive weights may be gathered into the matrix

$$\mathcal{W} = \begin{bmatrix} | & | & \cdots & | \\ \boldsymbol{\omega}_1 & \boldsymbol{\omega}_2 & \cdots & \boldsymbol{\omega}_{n_{\mathcal{M}}} \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{m_0 \times n_{\mathcal{M}}}, \quad (64)$$

whose columns correspond to the sampled manifold states and whose rows define discrete weight fields over the latent manifold.

The central objective of MAW-ECM is to identify the smallest possible common support capable of satisfying all local constraints simultaneously. Since the weights are constrained to remain nonnegative, the vector

$$\sum_{j=1}^{n_{\mathcal{M}}} \boldsymbol{\omega}_j$$

has a nonzero entry at position i if and only if the i -th candidate integration point receives a nonzero weight for at least one sampled manifold state. Therefore, the cardinality of the union of all local supports may be measured through the ℓ_0 pseudo-norm

$$\left\| \sum_{j=1}^{n_{\mathcal{M}}} \boldsymbol{\omega}_j \right\|_0. \quad (65)$$

This support-union interpretation closely parallels the formulation advocated by the authors in [6] in the context of subspace-adaptive weights ECM (SAW-ECM), which may be regarded as the subspace counterpart of the present manifold-adaptive construction. The main difference is that the discrete subspace index is here replaced by a sampled nonlinear manifold of latent states. Furthermore, unlike a finite collection of essentially independent subspaces, neighboring points on the latent manifold are expected to induce related local cubature rules, whose associated adaptive weights should evolve in a sufficiently smooth and coherent manner across the manifold. Consequently, besides promoting a sparse common support, it is also natural to penalize excessively irregular variations of the adaptive weights over the latent manifold, in the sense discussed in Item 4 of Section 7.1.

Based on the above considerations, we propose the following manifold-adaptive sparsification problem:

$$\begin{aligned} \min_{\omega_1, \dots, \omega_{n_{\mathcal{M}}}} \quad & \left\| \sum_{j=1}^{n_{\mathcal{M}}} \omega_j \right\|_0 + \mathcal{R}(\mathcal{W}) \\ \text{subject to} \quad & \mathbf{U}_j^T \omega_j = \mathbf{b}_j, \quad j = 1, \dots, n_{\mathcal{M}}, \\ & \omega_j \geq 0, \quad j = 1, \dots, n_{\mathcal{M}}. \end{aligned} \tag{66}$$

The first term in the objective function promotes a sparse common support shared across the sampled manifold states (that is, it seeks to minimize the total number of integration points), whereas the regularization functional $\mathcal{R}(\mathcal{W})$ penalizes highly irregular variations of the adaptive weights over the latent manifold. To put it differently, $\mathcal{R}(\mathcal{W})$ measures the distortion of the adaptive weight fields relative to the initial fixed-weight configuration, for which all sampled manifold states share the same cubature weights and thus $\mathcal{R}(\mathcal{W})$ vanishes. The parameter $\alpha \geq 0$ controls the strength of this regularization. It should be noted that, at this juncture of the discussion, $\mathcal{R}(\mathcal{W})$ is introduced only in abstract form. Specific choices associated with the constructive pruning strategy will be discussed later.

7.4 Pruning strategy

7.4.1 Overview

The formulation of Problem (66) is valuable at the conceptual level because it provides an explicit mathematical formulation of the manifold-adaptive sparsification objective underlying MAW-ECM. Nevertheless, much like the fixed-weight formulation in (34), the problem is not amenable to direct solution in practice because it belongs to the class of ℓ_0 -minimization problems, which are well known to be NP-hard [16]. In other words, the difficulty stems from the combinatorial nature of the ℓ_0 -term in the objective function.

Practical approaches for ℓ_0 -minimization problems typically rely either on suboptimal greedy procedures or on suitable convex relaxations, most commonly obtained by replacing the ℓ_0 -objective with an ℓ_1 -based surrogate promoting sparsity. For instance, the latter strategy is the philosophy underlying the Empirical Quadrature Method, EQM, introduced by Patera and co-workers [19, 20] to solve Problem (34) in alternative way to that of those based on nonnegative least-squares concept, as in the ECM and the ECSW.

The ℓ_1 -based convexification route was previously explored by the authors in [6] in the context of the aforementioned SAW-ECM (the sparsification problem in SAW-ECM is structurally analogous to Problem (66), but without the manifold-regularization term). The main conclusion drawn in [6] (and that applies for the problem at hand) is that the ℓ_1 -based convexification route is not viable in practice. The reason is twofold. Firstly, the LP formulation requires stacking all adaptive weight vectors into a single global optimization variable (in the present notation, this amounts to vectorizing the matrix $\mathcal{W}$ introduced in Eq. (64), leading to a variable of dimension $m_0 n_{\mathcal{M}}$). In doing so, the formulation loses sight of the fundamentally local nature of the cubature constraints associated with each sampled manifold state, while the computational cost rapidly becomes prohibitive as $n_{\mathcal{M}}$ grows. Secondly, even in relatively small problems for which the LP problem remains tractable, the numerical experiments reported in [6] showed that the resulting supports were systematically larger than those obtained with the final greedy SAW-ECM strategy.

Accordingly, the present work does not pursue this ℓ_1 -based route, but instead adopts a greedy iterative strategy that avoids treating the entire global optimization variable simultaneously. The underlying idea is inspired by the *pruning* strategies employed in the derivation of generalized quadrature rules [7, 18], where the goal is likewise to identify the smallest possible set of integration points, including both their positions and weights. This philosophy was adopted

by the authors in [12] to derive a continuous⁹ version of ECM.

The analogy with the present MAW–ECM setting is the following. In both cases, the starting point is an already feasible cubature rule possessing more points than strictly necessary. Sparsification is then achieved progressively by eliminating points that contribute the least to the satisfaction of the integration and positivity constraints, while continuously readjusting the remaining weights so that the constraints remain satisfied. In this sense, both approaches view sparsification not as a global combinatorial search over all admissible supports, but rather as a sequential pruning process in which integration points are removed one at a time from a common support while the remaining weights are redistributed to preserve local exactness conditions.

The essential difference lies in the nature of the design variables. In generalized quadrature rules [7, 18] and in CECM [12], both the positions and the weights of the cubature points are allowed to evolve continuously during the pruning process. In the present MAW–ECM setting, by contrast, the candidate points are fixed Gauss points inherited from the underlying finite element discretization, and only the manifold-dependent weights are modified. Moreover, in the present problem, the adaptive weights cannot evolve independently from one manifold state to another, but must vary in a sufficiently smooth and coherent manner over the latent manifold; enforcing such smoothness is precisely the role of the regularization term.

7.4.2 Pruning algorithm

8 Concluding remarks

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⁹The qualifier continuous refers to the fact that the positions of the cubature points are treated as continuous design variables, rather than as members of a discrete candidate set subject to selection.

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